

Formalizing Compactly Supported Stokes' Theorem in Lean 4 via Coordinate-Represented Integrals

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Abstract

We present a large-scale Lean 4 formalization of the local-to-global core of the compactly supported Stokes theorem for manifolds modeled on a half-space. The main public theorem is a coordinate-represented compact-support Stokes theorem: from chartwise smoothness of an n -form and compactness of the closure of its support, the development constructs a finite chart-box localization, support-controlled smooth refinements, zero-localized chart representatives, half-space local Stokes data, and canonical represented bulk and boundary endpoints, and proves that the two endpoints are equal.

The adjective “represented” is a precise statement about the theorem’s interface. The endpoints are the finite coordinate sums produced by the classical proof after choosing charts, boxes, and a partition of unity. They are not approximations, placeholders, or user-supplied certificates. The formalization proves the substantive Stokes proof payload at this represented level: compact support is turned into finite data, totalized chart representatives are replaced by zero-localized representatives, artificial box faces vanish by topological support control, the lower half-space face carries the outward-first boundary sign, and the local identities are reconstructed as global represented endpoints.

The active project contains 544 Lean files and about 1.65×10^5 lines under `Stokes/`. The preferred public entry point is the first-principles `representedStokes` theorem; its public fields are chartwise smoothness and compactness of the closed support. The artifact builds without `sorry`, `admit`, or project axioms, and the theorem-level axiom audit reports only the standard Lean/mathlib axioms `propext`, `Classical.choice`, and `Quot.sound`. A central engineering contribution is the passage from certificate-consuming theorem shapes to a certificate-generating one: finite covers, refinements, zero-support facts, local regularity fields, and endpoint reconstruction are not public hypotheses, but consequences constructed by the proof. What remains beyond the present claim is the comparison of the represented endpoints with a future or external mathlib-native manifold integration API.

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*Artifact repository: <https://github.com/evaristebernhard/Stokes>.

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1 Introduction

Stokes’ theorem is one of the places where the informal compression power of mathematics is most visible. In a textbook proof on a compactly supported form, one takes a finite chart cover, chooses a partition of unity, applies the Euclidean theorem on each coordinate box, observes that the artificial faces cancel or vanish, and identifies the remaining terms with the boundary integral [Spi65, Mun91, Lee13, Tu11]. The entire local-to-global passage may occupy a page. In a proof assistant, however, none of these choices is silent. The finite cover has an index type. The support of each partition coefficient has to be controlled. The chart representatives are total functions. Boundary charts are modeled on a half-space rather than on all of $\mathbb{R}^{n+1}$. The auxiliary faces of coordinate boxes must be separated from the true boundary face. The final finite sum must be reconstructed as a canonical global endpoint.

This paper presents a Lean 4 formalization of that local-to-global core for a coordinate-represented compact-support Stokes theorem, building on Lean’s dependent type theory and mathlib’s analysis, topology, and manifold infrastructure [dMKA⁺15, dMU21, Lea26, AdMKU26, The20, The26]. The main public theorem is the first-principles theorem named:

`CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.representedStokes`

Its public input consists of chartwise smoothness of the n -form and compactness of the closure of its support. From these two analytic hypotheses, under the ambient half-space model assumptions carried by the surrounding Lean context, the development constructs the hidden data of the proof: pointwise chart boxes, an open finite selected cover, a support-controlled partition, a boundary half-space box refinement, smooth refined coefficients, zero-localized chart representatives, local half-space Stokes fields, and two canonical represented endpoints. It then proves that the generated bulk endpoint equals the generated boundary endpoint.

The theorem is represented rather than native in a precise sense. The endpoints are not arbitrary real numbers supplied by a user. They are the finite coordinate sums produced by the classical proof after localizing the form and applying Stokes on finitely many half-space boxes. The theorem proves the equality

$$\text{BulkRep}(\omega) = \text{BoundaryRep}(\omega),$$

where both sides are generated by the formal construction. What is not claimed is a comparison theorem between these represented endpoints and a future mathlib-native API for integrating differential forms over oriented manifolds with boundary. That comparison is important, but it is a different layer: it identifies a canonical finite coordinate construction with a chosen native global integral. The present artifact proves the Stokes proof payload at the represented level.

1.1 What Lean forces us to expose

The traditional statement

$$\int_M d\omega = \int_{\partial M} \omega$$

suppresses several constructions. The formalization shows that these are not mere implementation details.

First, compact support is a constructive resource. The proof does not simply use compactness abstractly; it uses compactness to extract a finite family of chart boxes and to make every local summand subordinate to one of them. The development therefore works with a compact carrier

$$K = \overline{\text{supp}(\omega)}$$

and proves that the selected partition has topological support controlled over K . This support control is later used to kill artificial boundary faces.

Second, boundary charts are half-space charts. A coordinate box adjacent to the boundary is not an ordinary closed box sitting in an ambient open subset of $\mathbb{R}^{n+1}$. It touches the face $x_0 = 0$, and the local theorem must separate the open-box differentiability needed for the bulk term from the closed-box continuity and face regularity needed for boundary terms. This is why the development contains a genuine half-space local Stokes layer, centered around `HalfSpaceBoxInteriorStokesFields`, rather than pretending that boundary boxes have ordinary Euclidean neighborhoods across the boundary.

Third, Lean functions are total. A chart representative such as `transitionPullbackInChart` has values even outside the meaningful chart source, target, or overlap. A naive theorem saying that the support of this total representative is contained in a selected coordinate box is often the wrong theorem. The project introduces `ManifoldForm.transitionPullbackInChartZero`, a zero-localized chart representative, and proves support theorems for that representative. The key formal statement

`zeroSupportOfSelectedSmoothRefinement`

turns the informal phrase “the localized form is supported in the box” into a Lean-checkable topological support inclusion for the actual refined representatives used by the global proof.

Fourth, the endpoint reconstruction is itself mathematics. Once each local half-space box satisfies Stokes and the artificial faces vanish, one still has to prove that the nested finite sum over selected boundary charts and refined pieces is the canonical represented bulk endpoint, and similarly for the boundary endpoint. This is not a naming convention. It is a finite-sum reconstruction theorem built into the localized refined endpoint layer.

1.2 Scale of the artifact

The active project contains 544 Lean files under `Stokes/`, totaling about 1.65×10^5 lines. The size is not accidental. A large part of the work is the removal of theorem-facing certificates. Earlier intermediate routes exposed selected covers, refinement data, local Stokes records, endpoint adapters, target-image control, or support assumptions. These intermediate APIs were useful for stabilizing the proof, but the final public route compresses them behind first-principles input. The default public import has also been slimmed so that the preferred entry point is the first-principles represented theorem rather than the historical development scaffolding.

The theorem is audited in the usual Lean way. The repository contains no project-level uses of `sorry`, `admit`, or new axioms in the active Lean sources, and the theorem-level axiom audit for the first-principles statement reports the standard Lean/mathlib axioms

`propext, Classical.choice, Quot.sound.`

The paper records the commands used for this audit in [Sections C](#) and [10](#).

1.3 Certificate generation as a contribution

A useful way to read the development is as a transformation from certificate-consuming formalization to certificate-generating formalization. An early formal milestone for a theorem of this kind can be logically correct while still asking the user for proof objects that arise only inside the local-to-global argument: a selected finite cover, collar boxes, refined partition coefficients, local Stokes fields, support-in-box assumptions, image-control packages, and endpoint adapters. Such a theorem proves a real identity, but its interface is not yet the mathematical theorem one wants to state.

The final route continues past that milestone. Each former certificate field is either proved unnecessary or generated from earlier mathematical data. The zero-localized representative is the sharpest example. The tempting support assumption for the ordinary total chart representative is not merely inconvenient; it is the wrong shape of theorem. The proof replaces the representative by one that agrees on the chart source and is zero elsewhere, then proves the topological support statement required by half-space Stokes. This is a mathematical correction of the formal object, not a cosmetic API change.

1.4 Contributions

The main contributions are the following.

- A coordinate-represented compact-support Stokes theorem in Lean 4 whose public input is chartwise smoothness and compactness of the closed support.
- A formal local-to-global construction turning compact support into an open finite selected cover, a support-controlled partition, and a smooth boundary half-space refinement.
- A half-space local Stokes layer that separates open-interior differentiability, closed-box regularity, artificial-face vanishing, and the outward-first sign of the true boundary face.
- A zero-localization method for chart representatives, replacing false ordinary-support assumptions for totalized chart functions by generated topological-support theorems for the correct local representative.
- A canonical represented endpoint layer proving equality of generated finite coordinate bulk and boundary sums, rather than consuming those endpoints as external certificates.
- A proof-engineering compression from certificate-consuming intermediate routes to a certificate-generating first-principles public theorem.

1.5 Organization

[Section 2](#) explains represented endpoints and the exact claim of the theorem. [Section 3](#) gives the local-to-global proof decomposition from the perspective of formalization. The next four sections discuss the main mathematical bottlenecks: half-space local Stokes ([Section 5](#)), zero-localized support ([Section 6](#)), compact finite data ([Section 7](#)), and endpoint reconstruction ([Section 8](#)). [Section 9](#) states the main Lean theorem and describes the first-principles route. [Section 10](#) records artifact scale and verification. Related work and limitations appear in [Sections 11](#) and [12](#). The appendix gives a theorem and API map for readers navigating the Lean files.

2 The Represented Stokes Theorem

The central statement of this artifact is a represented Stokes theorem. The word *represented* is meant in the same mathematical spirit as a represented measure, a coordinate expression, or a chain-level representative: the theorem chooses a canonical finite coordinate expression produced by the proof and proves Stokes at that level. It is not a statement about approximation, and it is not a placeholder for a missing equality.

2.1 The classical finite-coordinate endpoint

Let ω be a compactly supported n -form on a manifold modeled on a half-space. The usual proof of Stokes begins by choosing a finite family of charts and a partition of unity (ρ_j) on the compact support. The bulk side is decomposed formally as

$$\int_M d\omega = \sum_j \int_M d(\rho_j \omega),$$

up to the standard partition and support arguments. For interior charts, the coordinate theorem reduces the local term to an ordinary box theorem whose boundary contributions are artificial and vanish by support. For boundary charts, each localized term is further refined into finitely many half-space boxes. On such a box $Q_{i,q}$, the local theorem gives a term of the form

$$B_{i,q}(\omega) = C_{i,q}(\omega),$$

where $B_{i,q}$ is the local bulk integral of the coordinate representative of a localized form and $C_{i,q}$ is the true lower-face boundary contribution with the half-space boundary sign.

The represented bulk endpoint is the nested finite sum of the $B_{i,q}$. The represented boundary endpoint is the nested finite sum of the $C_{i,q}$. In schematic form,

$$\text{BulkRep}(\omega) = \sum_i \sum_{q \in Q_i} B_{i,q}(\omega), \quad \text{BoundaryRep}(\omega) = \sum_i \sum_{q \in Q_i} C_{i,q}(\omega).$$

The theorem proves $\text{BulkRep}(\omega) = \text{BoundaryRep}(\omega)$ for the canonical finite data generated from compact support and chartwise smoothness.

Definition 2.1 (Represented Stokes route). *A represented Stokes route for a compactly supported form consists of finite boundary chart indices i , finite refined half-space pieces $q \in Q_i$, box corners $a_{i,q} \leq b_{i,q}$ with $(a_{i,q})_0 = 0$, localized coefficients $\rho_{i,q}$, and coordinate representatives $\alpha_{i,q}$ satisfying the following conditions.*

1. *The coefficients arise from a support-controlled partition and smooth refinement of the compact carrier.*
2. *The topological support of $\alpha_{i,q}$ is contained in the assigned half-space support box, so every artificial face of the coordinate box gives zero.*
3. *The representative $\alpha_{i,q}$ satisfies the open-interior and closed-box regularity assumptions needed for the half-space local theorem.*
4. *The local bulk and lower-face boundary terms assemble into generated finite endpoint sums.*

The represented endpoints of the route are the two finite sums produced by the last item.

The Lean construction makes this definition concrete. The endpoint layer stores the refined pieces and local terms in `CoverIndexedBoundaryLocalizedRefinedPartition`; the zero-localized certificate supplies the support condition; and the selected regularity constructor supplies the analytic fields. The first-principles theorem proves that the route exists canonically from the theorem hypotheses.

2.2 Endpoint algebra

The represented endpoint is best understood as a finite algebraic object. For each active pair (i, q) , the route assigns a localized coefficient $\rho_{i,q}$, a boundary chart x_i , and half-space box corners $(a_{i,q}, b_{i,q})$. The local bulk term has the form

$$B_{i,q} = \int_{[a_{i,q}, b_{i,q}]} \text{coeff}_{\text{top}}(d(\text{ZeroRep}_{x_i, x_i}(\rho_{i,q}\omega))),$$

where the integral notation abbreviates the coordinate box integral used by the formal local theorem. The boundary term is

$$C_{i,q} = \text{sgn}_{\partial\mathbb{H}} \int_{\{x_0=0\} \cap [a_{i,q}, b_{i,q}]} \text{coeff}_{\partial}(\text{ZeroRep}_{x_i, x_i}(\rho_{i,q}\omega)).$$

The represented endpoints are then exactly

$$\text{BulkRep}(\omega) = \sum_i \sum_{q \in Q_i} B_{i,q}, \quad \text{BoundaryRep}(\omega) = \sum_i \sum_{q \in Q_i} C_{i,q}.$$

This display is schematic, but it captures the formal structure: both endpoints use the same finite selected pieces, the same refined coefficients, and the same chart choices. The local Stokes theorem proves $B_{i,q} = C_{i,q}$ piecewise; endpoint reconstruction proves that summing these equalities gives the equality of the generated endpoint definitions.

This is exactly the equality obtained in the middle of the textbook proof, after localization and before comparison with a global manifold integration interface. In informal mathematics one usually identifies this finite coordinate expression with the left and right sides of the global theorem immediately. In Lean, that comparison should be a theorem about the chosen global integration API. The present work isolates and proves the coordinate Stokes content independently of that future comparison layer.

2.3 What is generated, and what is not supplied

The strongest evidence that the represented endpoints are mathematical rather than arbitrary is their construction. In the final route, the user does not provide the endpoints. The top-level structure

CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput

has two fields:

```
structure CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput
  (I : ModelWithCorners Real (Fin (n + 1) -> Real) H)
  (omega : ManifoldForm I M n) where
  chartwiseSmooth : ManifoldForm.ChartwiseSmooth I omega
  compactSupport : IsCompact (closure (ManifoldForm.support I omega))
```

The canonical bulk and boundary endpoints are definitions in the namespace of this structure:

```
def canonicalRepresentedBulkIntegral : Real := ...
def canonicalRepresentedBoundaryIntegral : Real := ...
```

They are obtained by first building an intrinsic input on the compact carrier $\overline{\text{supp}(\omega)}$, then following the canonical selected-cover and refinement route. The theorem is:

```
theorem representedStokes :
  X.canonicalRepresentedBulkIntegral (I := I) (omega := omega) =
  X.canonicalRepresentedBoundaryIntegral (I := I) (omega := omega)
```

This shape matters. A certificate-consuming theorem would have asked the user to provide a finite cover, local Stokes data, endpoint adapters, and finally the names of the two real numbers to compare. The first-principles theorem does the opposite. It internalizes the choices and exposes the generated endpoint names as projections of the proof route.

The noncomputable choices inside the proof do not weaken the statement. They are the ordinary choices already present in the classical proof: finite subcovers and subordinate partitions. The theorem fixes a construction path and proves the equality for the endpoints generated by that path. A future native-integral comparison theorem can then prove that these generated endpoints agree with a chart-independent integral, but the Stokes equality of the generated endpoints is already a theorem.

2.4 The represented-to-native boundary

The represented theorem should be read as a theorem at a stable mathematical boundary. It proves the finite coordinate equality produced by the classical proof. It does not yet assert:

$$\text{BulkRep}(\omega) = \int_M d\omega, \quad \text{BoundaryRep}(\omega) = \int_{\partial M} \omega$$

for a particular future or external native manifold integration API. Such a comparison requires a convention for integrating differential forms over oriented manifolds with boundary in mathlib, boundary orientation data, measure normalization, chart-independence theorems, and partition-independence theorems.

The distinction is useful rather than apologetic. It separates two logically different parts of Stokes:

- the local-to-global Stokes proof, where compact support, charts, partitions, half-space boxes, artificial faces, and signs are handled;
- the representation theorem identifying the resulting finite coordinate sums with a native manifold integral interface.

The artifact formalizes the first part at scale. The second part is the natural next interface once a suitable native integration API is available.

2.5 Why represented endpoints are canonical

There are several sources of noncomputability in the theorem: finite subcovers, partitions of unity, and choices of refinements. This does not make the endpoint arbitrary. In Lean, “canonical” here means canonical relative to the chosen construction path in the theorem namespace. The endpoint is defined by that path, and the theorem proves equality for those generated definitions. This is analogous to choosing a proof-generated representative in homological algebra and proving a chain-level equality before proving independence of representative.

The formal route fixes the carrier K , extracts a selected open cover from pointwise chart-box data, constructs a support-controlled partition, refines boundary active carriers, builds local zero-localized Stokes data, and then defines the endpoints by finite sums over the resulting boundary pieces. Thus the represented values are stable enough to state, audit, and use as the target of future native comparison theorems.

Theorem 2.2 (Represented local-to-global Stokes, mathematical reading). *Let ω be a chart-wise smooth compactly supported n -form in the half-space model context of the artifact. For the finite represented Stokes route generated from $\text{supp}(\omega)$, every refined local half-space term satisfies local Stokes, artificial coordinate faces vanish, and therefore the generated represented endpoints satisfy*

$$\text{BulkRep}(\omega) = \text{BoundaryRep}(\omega).$$

Definition 2.3 (Coordinate-represented compact-support Stokes). *In this paper, a coordinate-represented compact-support Stokes theorem is a Lean theorem which, from chartwise smoothness and compactness of the closed support, constructs finite coordinate data following the standard local proof and proves equality of the generated finite bulk and boundary coordinate endpoints.*

This definition is the claim made by the artifact. It is deliberately stronger than a local Euclidean theorem and deliberately narrower than a full mathlib-native manifold integral statement.

3 Local-to-Global Proof Decomposition

The formalization is best understood as proof archaeology: it takes a compact paper proof and recovers the hidden mathematical objects one by one. Each object becomes necessary because a later step needs a precise support, smoothness, or finite-sum theorem.

3.1 The compressed paper proof

A standard proof for compactly supported Stokes can be summarized as follows. Choose a compact carrier K containing the support of ω . Cover K by finitely many coordinate neighborhoods adapted to the boundary. Let (ρ_j) be a smooth partition of unity subordinate to that cover. Then write $\omega = \sum_j \rho_j \omega$ on K , apply local Stokes to each $\rho_j \omega$, use support to discard artificial faces, and sum the remaining boundary terms [Spi65, GP74, War83, Lee13, Mor01].

Every phrase in this paragraph hides a Lean obligation.

- “Cover by finitely many coordinate neighborhoods” requires pointwise chart-box data, open shrink data, compactness extraction, an index type, and membership theorems for the selected boxes.
- “Partition subordinate to the cover” requires not merely functions summing to one, but topological support control strong enough to survive localization and coordinate transition.
- “Apply local Stokes” requires a half-space local theorem, because boundary charts are not ordinary Euclidean charts.
- “Artificial faces vanish” requires a theorem relating topological support of a zero-localized chart representative to the half-space support box used in the local theorem.
- “Sum the remaining boundary terms” requires a finite-sum endpoint reconstruction theorem, because refined boundary pieces are indexed by nested finite data.

The project therefore decomposes the theorem into layers. Each layer solves a specific mathematical problem and exposes exactly the theorem needed by the next layer.

3.2 Layer 1: Euclidean and half-space local Stokes

The bottom layer is Euclidean. The project builds on mathlib rather than reimplementing analysis. Local box Stokes is organized around the divergence theorem and the coordinate form representation, following the classical measure-theoretic and multivariable-analysis background that also underlies formalized integration work in theorem provers [Rud87, EG92, Fed69, Gou22, vD21]. For boundary charts, however, the local model is the upper half-space

$$\mathbb{H}^{n+1} = \{x : \mathbb{R}^{n+1} \mid 0 \leq x_0\}.$$

The formal half-space theorem has to distinguish the true lower face $x_0 = 0$ from all other faces of the coordinate box.

The key record is `HalfSpaceBoxInteriorStokesFields`. It packages the data needed to apply the Euclidean box theorem without requiring the closed box to sit inside an ambient open subset across the boundary. The theorem

`halfSpaceLocalStokes_compactSupport_of_interiorFields`

then says that, under the support condition killing artificial faces, the half-space local bulk term equals the signed true boundary term.

3.3 Layer 2: Chartwise forms and total representatives

The next layer connects manifold forms to coordinate forms. A chart representative in Lean is a total function on the coordinate space. Totality is convenient for functions but dangerous for support statements. The informal support assertion

$$\text{supp}(\varphi_*(\rho\omega)) \subseteq Q$$

means that the local representative is supported in Q where the chart is valid. Lean must assign values outside the valid region too. If those values are artifacts of totalization, the ordinary support theorem is not the theorem the proof needs.

The formal solution is the zero-localized representative

`ManifoldForm.transitionPullbackInChartZero`.

It agrees with the usual transition representative on the chart source and is zero outside. The project proves source, target, and overlap support theorems for it, and then proves that localized forms constructed by the selected smooth refinement have topological support inside their assigned half-space boxes.

3.4 Layer 3: Compactness to finite selected data

Compactness enters constructively. The first-principles theorem takes

$$K = \overline{\text{supp}(\omega)}$$

and uses compactness of K . Pointwise chart-box data is generated from the half-space model and self charts. From pointwise neighborhoods, the formalization chooses open shrinks inside chart boxes, extracts a finite selected cover, and constructs a support-controlled partition.

The phrase support-controlled is essential. The partition is not used merely to sum to one. Its topological support, intersected with K , must remain inside the assigned selected open set and ultimately inside the assigned coordinate box. Later support theorems for localized forms depend on this exact statement.

3.5 Layer 4: Boundary refinement

The boundary part requires a second finite refinement. A selected boundary chart may contain an active carrier that is too large for a single half-space box. The formal route refines each boundary active carrier into finitely many half-space boxes and constructs smooth subpartition coefficients. The resulting local pieces are indexed by a pair (i, q) , where i is a boundary chart index and q is a refined boundary piece.

This is one reason the endpoint layer cannot be a one-box-per-chart theorem. The final represented endpoint is a nested finite sum, and the formalization must keep track of the equality between the generated refined pieces and the selected boundary finite cover.



Figure 1: The first-principles route. The public hypotheses generate the compact carrier, selected cover, support-controlled partition, smooth refinement, zero-localized support, local fields, and represented endpoints. In this and subsequent dependency figures, light-green rectangular nodes denote definitions or constructed data, while dark-green oval nodes denote theorem-level statements or derived equalities.

3.6 Layer 5: Local Stokes data to endpoint equality

After localization and refinement, each piece has:

- a half-space box with lower normal coordinate zero;
- a zero-localized chart representative supported in that box;
- analytic regularity fields sufficient for local Stokes;
- a local bulk term and a local true-boundary term.

The local theorem gives equality for each piece. Finite summation gives equality of nested sums. The endpoint layer identifies those nested sums with the generated represented bulk and boundary endpoints.

This decomposition explains the final theorem’s shape. Its proof is short because each hidden mathematical step has already been turned into a named constructor or theorem. The short top-level proof is the result of compression, not evidence that the theorem is a thin statement.

3.7 A useful invariant

A recurring invariant in the development is the following:

Every local summand must carry enough support information to kill the faces that are not part of the manifold boundary.

Many intermediate certificate fields existed because this invariant had not yet been derived from earlier data. The final first-principles route proves the invariant from compact support, selected open support control, and zero-localized chart representatives. This is the mathematical reason the public API can be small.

4 Formalization Obstacles as Mathematical Lemmas

The development repeatedly encountered situations where a sentence that is acceptable in a paper proof was not a well-formed theorem in Lean. The useful lesson is not that Lean is fussy. Rather, the formalization identifies the mathematical lemma that the sentence was abbreviating. This section collects five such obstacles. They are the main reason the final theorem is stronger than a certificate-consuming assembly theorem: each obstacle was eventually turned into a constructor or theorem generated from compact support, chartwise smoothness, and half-space geometry.

Paper phrase	Ill-shaped Lean theorem	Corrected theorem/API	Role in Stokes
“smooth on the boundary box”	ambient smoothness on an open neighborhood of a closed box touching $x_0 = 0$	interior Stokes fields and open-interior/closed-box regularity	legal boundary local Stokes
“the chart representative is supported in the box”	support of the totalized chart representative	zero-localized representative and selected zero-support theorem	support theorem for the correct representative
“the artificial faces vanish”	face integrals are zero by informal inspection	face-coefficient support predicate and artificial-face remainder theorems	removes all non-boundary box faces
“choose finitely many charts”	an arbitrary finite cover disconnected from partition support	open shrinks, selected finite covers, and support-controlled partitions	turns compact support into usable local data
“sum the local identities”	local sums and endpoint definitions use different finite indices	generated represented endpoints over the same refined pieces	global represented equality

Table 1: Five paper-level phrases and the precise theorem shapes needed in Lean. The final first-principles theorem is small because the corrected theorems are generated internally rather than exposed as public hypotheses.

Table 1 is the organizing principle for this section. Each row is an instance of the same pattern. The informal phrase is not false in ordinary mathematical practice; it is a compressed reference to a more precise lemma. A certificate-consuming formalization can stop by asking the user to provide that lemma. The contribution of the final route is that the lemma is proved from the preceding data and no longer appears as a theorem-facing field.

4.1 Boundary smoothness is not ambient closed-box smoothness

The informal local proof near a boundary point often says that a coordinate representative is smooth on a box. If the model is a half-space, the word “box” hides a choice. The closed coordinate box may touch the boundary face $x_0 = 0$. Requiring an ambient open neighborhood of the closed box inside the chart target is then too strong: it asks for Euclidean data across the boundary, where the boundary chart is not intended to provide it.

The corrected mathematical lemma separates the roles of the box:

- differentiability for the bulk term is needed on the open interior $\prod_i (a_i, b_i)$;
- continuity and integrability are needed on the closed box and its faces;
- the lower normal face $x_0 = 0$ is allowed to lie on the model boundary.

The Lean record `HalfSpaceBoxInteriorStokesFields` is exactly this separation. It is not a bureaucratic certificate; it is the local theorem shape that avoids an invalid boundary-chart smoothness hypothesis.

The corrected theorem is not weaker; it is better aligned with the analysis. It asks for differentiability exactly where the bulk derivative is evaluated and for closed-box regularity exactly where the face integrals and measure-theoretic bridges need it. This is the difference between assuming a Euclidean extension across the boundary and proving a half-space theorem.

4.2 Support in a chart is not support of a total function

The phrase “the chart representative is supported in the box” normally means that the representative is supported in the box where the chart expression is meaningful. Lean functions are total, so the ordinary representative has values outside the meaningful chart region. If a theorem quantifies over the ordinary support of that total function, it may be asking for facts about values introduced only to make the function total.

The corrected lemma changes the representative. This is stronger than adding an auxiliary hypothesis to the old theorem. The old theorem speaks about a total function whose off-source values are artifacts of totalization; the new theorem speaks about the local representative intended by the proof. The development defines a zero-localized representative which agrees with the ordinary representative on the chart source and is zero outside. The support theorem is then a genuine topological statement:

$$\text{tsupp}(\text{ZeroRep}_{x,x}(\rho\omega)) \subseteq Q(a,b).$$

The Lean theorem `zeroSupportOfSelectedSmoothRefinement` proves this statement for the actual coefficients generated by the selected smooth refinement.

This is one of the main places where the formalization discovers a better mathematical interface. Asking the user for a support assumption on `transitionPullbackInChart` would have hidden the problem. Proving `zeroSupportOfSelectedSmoothRefinement` removes the assumption by proving that the generated representative has the support property that the informal proof silently uses.

The theorem therefore has two parts that must travel together. First, the zero-localized representative agrees with the ordinary representative on the source where the local integral is evaluated. Second, it has global support properties because it is zero outside the source. Without the agreement theorem the proof would be about the wrong local form; without the support theorem artificial-face cancellation would still be a public assumption.

4.3 Artificial faces vanish by support, not by notation

When Stokes is applied on a coordinate box, every face of the box appears. In a boundary chart only the lower normal face is the manifold boundary. The other faces are artifacts of the localization. A paper proof may say that they vanish because the support is inside the box. Lean requires the exact disjointness statement for each face coefficient and each artificial face.

The corrected lemma has the following structure:

$$\text{tsupp}(\alpha) \subseteq Q(a,b) \implies \text{every artificial face integral of } \alpha \text{ is } 0.$$

This is implemented through the face-support API in the half-space layer. The global proof does not assume artificial-face cancellation; it derives it from the zero-localized support theorem for each refined local representative.

The important point is that the support statement is made for face coefficients, not only for the form as an abstract object. The box theorem integrates coefficients on coordinate faces. The formalization proves that evaluating the form on a fixed face frame does not enlarge topological support, then uses the half-space support region to show that the artificial face set is disjoint from those coefficient supports.

4.4 Compactness is a construction principle

Compact support is not merely a hypothesis that allows one to write a finite sum. The proof uses it to construct the finite index type, the selected charts, the open shrinks, the partition coefficients, and the boundary refinement. If any of these choices is made independently, later support and endpoint reconstruction statements can fail to line up.

The corrected lemma is a compactness-to-data theorem: compactness of K together with pointwise chart-box neighborhoods produces a finite selected cover with a support-controlled partition. The selected cover remembers both the open sets used for the partition and the closed or half-space boxes used for support. This dual memory is what lets the local support facts feed into half-space Stokes.

Thus compactness is used to generate an indexed object with invariants, not only to prove existence of finitely many open sets. The finite selected data must remember enough information for three later consumers: the partition of unity, the half-space local theorem, and the endpoint summation.

4.5 Represented endpoints are finite coordinate theorems

The last hidden step is endpoint reconstruction. Once each local piece satisfies Stokes, the proof still has to show that the finite sum of local bulk terms is the generated represented bulk endpoint, and similarly for the boundary terms. This is not a definition-free conclusion: the local pieces are sigma-indexed by boundary chart and refined box data.

The corrected theorem keeps the endpoint generated from the same finite data used by the local proof. This prevents a common mismatch: proving local equalities over one index family and defining the global endpoint over a slightly different one. In the final route, the represented endpoints are canonical because they are induced by the selected finite route itself.

This is why the represented endpoint layer is a mathematical layer rather than a naming layer. The endpoint is not chosen after the local proof; it is the finite sum whose index family is shared by the cover, the refinement, the zero-support theorem, and the local Stokes theorem.

4.6 Summary of the compression

Each obstacle has the same final form:

$$\text{informal paper step} \rightsquigarrow \text{precise generated Lean theorem.}$$

The final first-principles API is small because these generated theorems replace the large certificates that earlier routes exposed. This is the central mathematical engineering pattern of the project.

5 Half-Space Local Stokes

The boundary case is the first place where the formal proof diverges sharply from an ordinary Euclidean box proof. A boundary chart has image in a half-space, not in all of $\mathbb{R}^{n+1}$. A local box may touch the boundary face $x_0 = 0$. The correct local theorem therefore has to see the half-space geometry, as in classical treatments of manifolds with boundary and differential forms [War83, AMR88, Lan95, Lee13].

5.1 What the paper proof omits

In the paper proof, after choosing a boundary chart one often writes a local coordinate box and applies Stokes. The intended picture is clear: the lower normal face is the actual boundary of the manifold, while the other faces are introduced by the coordinate box and should not contribute if the localized form is supported away from them. In a proof assistant, this picture becomes a family of statements:

- the lower normal face is the face with coordinate $x_0 = a_0 = 0$;
- its sign agrees with the outward-normal-first boundary convention;
- the upper normal face and the tangential faces are artificial;
- the artificial face integrals vanish under a topological support condition;
- the local analytic assumptions are valid for a box that may touch the half-space boundary.

The last point is especially important. It is tempting to require smoothness on an ambient open neighborhood of the closed box. That requirement is natural for interior boxes, but it is not the right boundary hypothesis. If the box touches $x_0 = 0$, there need not be a chart-target open set in the ambient Euclidean space containing the closed box on both sides of the boundary. The bulk derivative is needed on the open box interior; the boundary faces need continuity and integrability data. The formal theorem separates these assumptions instead of smuggling in an invalid ambient-open condition.

The false theorem shape is therefore:

$$\text{closed half-space box } [a, b] \subseteq U, \quad U \text{ ambient open, } U \subseteq \text{chart target.}$$

For a boundary box with $a_0 = 0$, this asks the chart target to contain Euclidean points with negative normal coordinate near the lower face. That is not part of the half-space chart. The correct statement replaces this single ambient-open condition by separate obligations: open-interior differentiability for the bulk term, closed-box coefficient regularity for face integrals, and support control for artificial faces.

5.2 The half-space primitives

The half-space layer defines the model set, boundary inclusion, boundary tangent frame, outward normal, and sign convention. The relevant names include `upperHalfSpace`, `boundaryInclusion`, `outwardFirstBoundaryFrame`, and `halfSpaceBoundarySign`. The theorem

$$\text{halfSpaceBoundarySign_eq_outwardFirstBoundaryOrientationSign}$$

aligns the coordinate lower-face sign with the outward-first convention used for the boundary. Without this bridge, the local lower-face term would remain a cube face term rather than the represented boundary contribution.

The support box is encoded by `halfSpaceSupportBox`. It is the portion of the coordinate box that is relevant for the half-space theorem. The formal support condition used by local Stokes is expressed through

$$\text{boxFaceCoeffTSupportInHalfSpaceBox.}$$

This condition is later obtained from the zero-localized support theorem for the selected smooth refinement.

Geometrically, `halfSpaceSupportBox` is asymmetric. In the normal coordinate it allows contact with the lower face $x_0 = 0$, because that is the true boundary. In the upper normal direction and in all tangential face directions it is strict, because those faces are artificial. This is why the support theorem can both retain the boundary contribution and remove the rest of the cube boundary.

Theorem 5.1 (Half-space compact-support local Stokes). *Let α be an n -form-valued coordinate representative on $\mathbb{R}^{n+1}$, and let $a \leq b$ be box corners with $a_0 = 0$. Suppose the interior Stokes fields for (α, a, b) are available and the topological support of α is contained in the corresponding half-space support box. Then*

$$\text{Bulk}_{a,b}(\alpha) = \text{sgn}_{\partial\mathbb{H}} \text{Boundary}_{a,b}^{x_0=0}(\alpha).$$

In Lean this is the theorem `halfSpaceLocalStokes_compactSupport_of_interiorFields`.

This theorem is the local mathematical engine used by the global route. The left-hand side is the coordinate bulk integral of the exterior derivative coefficient. The right-hand side is not an arbitrary selected face: it is the lower normal face, transported through the half-space boundary sign.

5.3 Artificial faces

The local Euclidean box theorem produces all faces of the box. For a boundary chart, only the lower normal face corresponds to the boundary of the manifold. The remaining terms are collected in

$$\text{halfSpaceLocalBoundaryRemainder}.$$

The main cancellation theorem has the schematic form

$$\text{tsupp}(\alpha) \subseteq Q(a, b) \implies R_{a,b}(\alpha) = 0.$$

In Lean this is represented by the chain:

```
boxRemainingFormFaceTerms_eq_zero_of_tsupport_subset_halfSpaceSupportBox
halfSpaceLocalBoundaryRemainder_eq_zero_of_tsupport_subset_halfSpaceSupportBox
```

These theorems are where the informal phrase “the support misses the artificial boundary” becomes a checkable proof. They also explain why the global theorem needs topological support control, not merely pointwise ordinary support information.

The proof has a simple mathematical shape. Euclidean box Stokes first writes the bulk integral as the sum of all signed face integrals. The half-space boundary API identifies the lower normal face with the true boundary term. For every other face, support containment in the half-space support box gives disjointness between the face coefficient support and that face, so the corresponding face integral is zero. The remaining equality is exactly [Theorem 5.1](#).

5.4 Proof sketch of the local theorem

The local half-space theorem is obtained by factoring the usual box theorem through three comparison steps.

Step 1: ordinary box Stokes. For a coordinate n -form α on $\mathbb{R}^{n+1}$, the Euclidean box theorem expresses the integral of the coordinate exterior derivative over $[a, b]$ as a signed sum over all coordinate faces. This is the analytic part of the proof closest to standard treatments of integration and geometric measure theory [[Mun91](#), [Rud87](#), [Fed69](#), [EG92](#)]. In the Lean development, this is the part delegated to the box-Stokes infrastructure. The `HalfSpaceBoxInteriorStokesFields` record stores exactly the hypotheses that this theorem consumes: continuity of signed coefficients on the closed box, differentiability on the open box away from a countable exceptional set, integrability of the divergence term, and the equality between the exterior derivative coefficient and the coordinate divergence integral.

Step 2: identify the lower normal face. Because $a_0 = 0$, the lower face in the normal coordinate is the model boundary of the half-space. The boundary integral layer proves that the coordinate lower-face term is the half-space boundary form integral, multiplied by the sign prescribed by the outward-first convention. This is the content of the lower-face and `halfSpaceBoundarySign` bridge lemmas.

Step 3: erase artificial faces. All remaining faces are introduced by the coordinate box. The support box is chosen so that a form whose topological support is contained in it has zero face coefficient on those artificial faces. Hence the remainder sum is zero. This is where the global zero-support theorem is consumed by the local layer.

The result is a local theorem whose hypotheses match what boundary charts can actually provide. It does not require a closed box to have an ambient open neighborhood across the boundary.

Local ingredient	Why it is needed	Lean location
open-interior differentiability	identifies the bulk derivative on the box interior	open-interior fields
closed-box continuity and integrability	makes the box and face integrals legal	interior Stokes fields
lower normal face sign	turns the cube face into the true boundary term	boundary-sign lemmas
topological support in the half-space support region	removes every artificial face	face-support predicate

5.5 Interior fields

The record

```

structure HalfSpaceBoxInteriorStokesFields
  (omega : (Fin (n + 1) -> Real) ->
    (Fin (n + 1) -> Real) [/^\^Fin n]->L[Real] Real)
  (a b : Fin (n + 1) -> Real) where
  le : a <= b
  lower_zero : a 0 = 0
  exceptional : Set (Fin (n + 1) -> Real)
  exceptional_countable : exceptional.Countable
  continuous_signedCoeff : ...
  hasFDerivAt_signedCoeff : ...
  integrable_divergence : ...
  bulk_eq_boxIntegral : ...

```

packages exactly what the half-space local theorem needs from the Euclidean box theorem. The exceptional set and differentiability fields match the interface of the underlying box-divergence theorem. The field `bulk_eq_boxIntegral` bridges the mathlib exterior derivative integrand and the coordinate divergence integrand used by the box theorem.

This record is the point where the formalization makes a mathematical distinction that a paper proof can blur. There is no single phrase “smooth on the box” that supplies all of these facts in the boundary case. Some facts live on the open interior, some on the closed box, and some are a.e. comparisons between two integrands. Bundling them into `HalfSpaceBoxInteriorStokesFields` gives the exact contract consumed by local Stokes, after which later constructors prove the fields automatically for the selected refined representatives.

The theorem

```

theorem halfSpaceLocalStokes_compactSupport_of_interiorFields

```

then turns these fields plus artificial-face support control into the local compact-support half-space Stokes identity. The version

```

HalfSpaceBoxInteriorStokesFields.withRemainder

```

keeps the artificial-face remainder visible, while

```

HalfSpaceBoxInteriorStokesFields.compactSupport

```

uses support containment to remove it.

5.6 From chartwise smoothness to local fields

The first-principles theorem does not ask the user for `HalfSpaceBoxInteriorStokesFields`. Those fields are generated by:

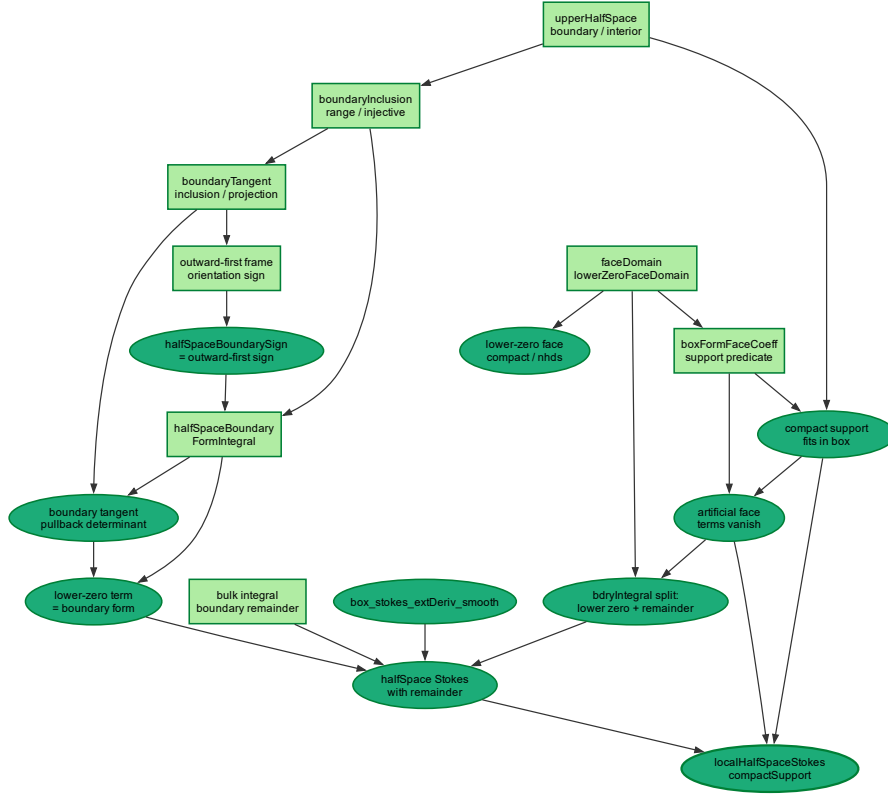


Figure 2: The half-space layer separates the true lower boundary face from artificial coordinate faces, and separates open-interior differentiability from closed-box face regularity. Light-green rectangular nodes denote definitions or constructed data; dark-green oval nodes denote theorem-level statements or derived equalities.

```
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.
interiorFieldsOfSelectedSmoothRefinement
```

The constructor uses chartwise smoothness of the original form, smoothness of the selected refined coefficients, and the fact that the selected closed boxes lie in the self chart transition source. It constructs open-interior smoothness fields and then applies

```
HalfSpaceBoxInteriorStokesFields.ofOpenInteriorSmoothness.
```

The mathematical point is that closed-box regularity is proved for the actual localized coordinate form on the selected box, while bulk differentiability is used on the open box interior. This prevents the global theorem from depending on a false boundary-chart ambient-open hypothesis.

This split is also important for degenerate boxes. Some coordinate intervals may collapse during finite refinement. In such cases, the closed-box regularity and integrability statements remain meaningful, while the open interior may be empty or lower dimensional. The local fields isolate the measure-theoretic bridge needed by the box theorem so that degenerate and nondegenerate cases can be handled uniformly by the surrounding constructors.

5.7 Why this layer is substantive

It would be possible to state a global theorem assuming a local Stokes theorem for every selected box. The artifact does more: it builds the local theorem from the half-space box infrastructure and then connects it to the selected refined representatives. This is why the final theorem can hide local Stokes fields. The local half-space content is not absent from the public API; it is proved and generated internally.

6 Support and Zero-Localization

The most deceptive obstacle in the formalization is support. Mathematicians habitually speak about a chart representative only where the chart is defined. Lean does not. A chart representative is a total function, and its support is a global statement about that total function. The central discovery of this part of the development is that the problem cannot be solved by asking for a stronger support hypothesis on the same representative. The representative itself has to be changed.

6.1 The false support theorem

Suppose ρ is a partition coefficient and ω is a form. A paper proof may say that the coordinate representative of $\rho\omega$ is supported in a coordinate box Q . In Lean, the corresponding ordinary representative is

`ManifoldForm.transitionPullbackInChart.`

It has a value at every coordinate point. Outside the chart source, target, or overlap, those values are artifacts of totalization. Thus a theorem of the form

$$\text{tsupp}(\text{transitionPullbackInChart}(\rho\omega)) \subseteq Q$$

may be too strong or simply unrelated to the informal argument. It asks Lean to control the chart representative even at points where the coordinate description should not be considered part of the local proof.

This problem explains several early certificate fields in the development. It is easy to request a support assumption from the user. It is harder, and more mathematically faithful, to prove the right theorem from compact support and partition support control. The right theorem is not about the unrestricted total representative.

One can view the issue as a small counterexample pattern. Suppose a chart expression is mathematically intended only on a set U , but Lean represents it by some total function $f : E \rightarrow V$. Even if f is perfectly controlled on U , its values on $E \setminus U$ may enlarge $\text{support}(f)$. A theorem about $\text{support}(f)$ is therefore not the same as a theorem about the local representative on U . The zero-localized representative replaces f by

$$y \mapsto \begin{cases} f(y), & y \in U, \\ 0, & y \notin U, \end{cases}$$

so that global support again expresses the intended local support. This is a change of object, not a notational convenience.

Remark 6.1 (The false support theorem). *The tempting theorem is:*

$$\text{tsupp}(\text{transitionPullbackInChart}(\rho\omega)) \subseteq Q.$$

The formalization treats this as the wrong target. The values of `transitionPullbackInChart` outside the transition source are present because Lean functions are total, not because the local

coordinate proof has mathematical content there. Strengthening the hypotheses to control those values would make the final theorem depend on an artifact of representation. The corrected target is the same inclusion for the zero-localized representative, together with an agreement theorem on the transition source.

6.2 Zero outside the chart domain

The project introduces a zero-localized representative:

`ManifoldForm.transitionPullbackInChartZero.`

It agrees with `transitionPullbackInChart` on the relevant chart source and is zero outside. The basic API proves:

- support is contained in the source;
- support is contained in the target;
- support is contained in the overlap;
- the zero representative agrees with the ordinary representative on the chart source.

The core names are:

```
transitionPullbackInChartZero_support_subset_source
transitionPullbackInChartZero_support_subset_target
transitionPullbackInChartZero_support_subset_overlap
transitionPullbackInChartZero_eqOn_transitionPullbackInChart
```

This changes the meaning of support control. Instead of proving that a totalized chart function behaves well outside its mathematical domain, the proof makes the representative zero there by definition and proves support properties for the resulting function. The point is not simply that zero extension is available; it is that the zero extension is the formal object corresponding to the informal phrase “the local chart representative”.

The equality theorem is as important as the support theorem. The zero representative must agree with the ordinary representative on the region where local integration is performed; otherwise the theorem would prove Stokes for a different object. This is why the API includes both support inclusions and agreement theorems such as `transitionPullbackInChartZero_eqOn_transitionPullbackInChart`.

The zero-localization API therefore has a three-part mathematical shape:

definition	make the off-source representative zero,
support theorem	prove source, target, and overlap support,
agreement theorem	recover the ordinary chart expression on the source.

All three are needed. The definition alone would give a different representative; the agreement theorem makes it harmless for integration on the selected box; the support theorem makes artificial-face cancellation available.

6.3 Localized forms

The global proof uses localized forms $\rho\omega$. Zero-localization has to interact with localization. The theorem

`ManifoldForm.transitionPullbackInChartZero_localizedForm`

rewrites the zero-localized representative of a localized form as the product of the coefficient representative and the zero-localized representative of the base form. From this, the project proves support inclusions such as

$$\text{transitionPullbackInChartZero_localizedForm_tsupport_subset_coefficient}$$

and

$$\text{transitionPullbackInChartZero_localizedForm_tsupport_subset_form}.$$

The decisive bridge is:

$\text{transitionPullbackInChartZero_localizedForm_tsupport_subset_halfSpaceSupportBox_of_support_subset_K}$
--

It formalizes the following argument:

$$\frac{\begin{array}{l} \text{supp}(\omega) \subseteq K, \\ \text{tsupp}(\rho) \cap K \text{ is controlled by the selected chart box,} \\ \text{the coordinate image of the active carrier is controlled,} \end{array}}{\text{tsupp}(\text{zero-localized representative of } \rho\omega) \subseteq \text{selected half-space support box}.}$$

This is the support theorem the half-space local Stokes layer needs.

Lemma 6.2 (Zero-localized support bridge). *Let K be a compact carrier containing $\text{supp}(\omega)$. Let ρ be a localization coefficient whose topological support on K is contained in a coordinate carrier, and suppose that the coordinate image of that carrier is contained in a half-space support box $Q(a, b)$. Then the zero-localized chart representative of $\rho\omega$ has topological support contained in $Q(a, b)$.*

This lemma is the formal replacement for the informal statement that “ $\rho\omega$ is supported in the selected box.” The latter statement is ambiguous in a total-function setting; the former is a precise theorem about the representative actually used by the local Stokes theorem.

At the level of Lean proof dependencies, the bridge decomposes as follows:

$$\frac{\begin{array}{l} \text{tsupp}(\text{ZeroRep}(\rho\omega)) \subseteq \text{tsupp}(\rho \text{ in chart}), \\ \text{tsupp}(\text{ZeroRep}(\rho\omega)) \subseteq \text{tsupp}(\text{ZeroRep}(\omega)), \\ \text{tsupp}(\text{ZeroRep}(\omega)) \subseteq \text{chartImage}(K), \\ \text{tsupp}(\rho \text{ in chart}) \cap \text{chartImage}(K) \subseteq Q(a, b), \end{array}}{\text{tsupp}(\text{ZeroRep}(\rho\omega)) \subseteq Q(a, b).}$$

The first two inclusions are product-support facts. The third is the global compact-support input transported through the zero representative. The last line is the selected-refinement support control. Their combination is exactly the support bridge displayed above.

There are two support operations in the proof. Ordinary support is algebraic: it records where a function is nonzero. Topological support is the closure of ordinary support and is the right object for face-disjointness and vanishing integrals. The development therefore does not stop at ordinary nonzero-set control. The final local Stokes input is a tsupp -level inclusion into a selected half-space support region. That region is designed to allow contact with the true boundary face $x_0 = 0$ while strictly avoiding the artificial faces of the coordinate box.

6.4 Selected smooth refinement

The final route applies the preceding bridge to the canonical smooth refinement selected by the first-principles theorem. The theorem is:

`CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.`
`zeroSupportOfSelectedSmoothRefinement`

states that for every selected boundary index i and active refined half-space piece q , the topological support of the zero-localized representative of the refined localized form is contained in the selected half-space support box:

$$\text{tsupp}(\text{ZeroRep}(\rho_{i,q}\omega)) \subseteq \text{halfSpaceSupportBox}(a_{i,q}, b_{i,q}).$$

Here $\rho_{i,q}$ is not a user-supplied coefficient. It is the coefficient generated by the selected smooth refinement.

This point is crucial for the first-principles theorem. If $\rho_{i,q}$ were arbitrary, the zero-support theorem would be false. It is true because $\rho_{i,q}$ is built from the selected boundary partition and the selected smooth subpartition, both of which carry support-control data from the compactness construction.

The proof uses a two-stage localization identity. The refined coefficient is the product of the selected boundary partition coefficient and the smooth subpartition coefficient. The proof rewrites

$$\rho_{i,q}\omega = \theta_{i,q}(\rho_i\omega)$$

and then uses the support of $\rho_i\omega$ to place the form in the boundary active carrier, while the support of $\theta_{i,q}$ places the coordinate representative inside the refined half-space box.

Proof idea for the selected theorem. The proof first shows that the boundary-localized form $\rho_i\omega$ is supported in the boundary active carrier. This uses the algebraic fact that support of a localized form is contained in the support of the coefficient and in the support of the original form, together with $\text{supp}(\omega) \subseteq K$. Next, the smooth subpartition coefficient $\theta_{i,q}$ is known, on that active carrier, to be supported inside the selected refined half-space box. The zero-localized product formula transfers these two support facts to the chart representative. The proof is carried out at the level of topological support for the coefficient and the zero base representative, using the product-support inclusions

$$\text{tsupp}(fg) \subseteq \text{tsupp}(f), \quad \text{tsupp}(fg) \subseteq \text{tsupp}(g).$$

Thus the theorem delivered to the half-space layer is already the required tsupp -inclusion; it is not a fragile ordinary-support statement later patched by hand. $\square$

6.5 Ordinary support versus topological support

Another formal distinction is ordinary support versus topological support. Many algebraic arguments first show that if a function is nonzero at a point, then that point lies in a desired set. But artificial-face vanishing is a closed-support statement. The theorem must control

$$\text{tsupp}(f) = \overline{\{x \mid f(x) \neq 0\}}.$$

In the half-space box proof the target support region is not a naive closed box. The set `halfSpaceSupportBox`(a, b) allows the support to meet the true lower normal face, but uses strict inequalities in the directions corresponding to artificial faces. This is exactly the geometry needed by the face-cancellation theorem: it proves that the artificial face set is disjoint

from the topological support. Consequently the proof is organized to produce `tsupp`-statements directly for the localized coefficients and zero-localized representatives.

Closed-set upgrades still appear in the surrounding support infrastructure in the familiar form

$$\text{support}(f) \subseteq C, \quad C \text{ closed} \implies \text{tsupp}(f) \subseteq C.$$

But the final half-space input is more specific: it is a topological-support inclusion into a region chosen so that artificial faces are avoided and the true boundary face is retained. This is why the support theorem is not merely “the coefficient is zero on the face.” It is a coordinated package of zero-extension, product support, compact carrier support, and half-space box geometry.

6.6 Contribution to the global theorem

Zero-localized support is the bridge between compact finite data and local Stokes. The support-controlled partition and smooth refinement produce coefficient support information. Zero-localization turns it into support information for the coordinate form representative. The half-space layer then uses that support information to remove artificial faces. Without this bridge, the final theorem would have to expose an unnatural support assumption for ordinary total chart representatives. With it, the first-principles theorem requires no such field.

7 Compact Support to Finite Data

Compact support is the theorem’s main global hypothesis. In the proof, it is not used as a passive assumption. It is the mechanism that turns pointwise chart geometry into a finite coordinate calculation. This is the constructive content of the compact-support proof found in classical differential-form treatments, where compactness, partitions of unity, and local coordinate Stokes are used together [Mun91, BT82, Mor01, Lee13].

7.1 The compact carrier

The first-principles theorem fixes the carrier

$$K = \overline{\text{supp}(\omega)}.$$

This choice is made by the definition

```
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.supportCarrier.
```

The public compactness field is

```
IsCompact (closure (ManifoldForm.support I omega)).
```

The support inclusion needed by the intrinsic route is then simply $\text{supp}(\omega) \subseteq \overline{\text{supp}(\omega)}$, discharged internally by `subset_closure`.

This design matters because it removes an otherwise arbitrary carrier from the public theorem. Earlier routes allowed a user to choose a compact set K and prove $\text{supp}(\omega) \subseteq K$. That is mathematically reasonable, but the first-principles form expresses the standard compact-support hypothesis directly: the closure of the support is compact.

This choice also fixes the direction of later support proofs. Every localized summand is ultimately controlled by showing that its nonzero points come from nonzero points of ω , hence lie in K . The theorem never needs the user to guess a larger carrier with additional geometric properties. All geometry is generated from the canonical carrier.

This is a small but important interface decision. A theorem with an arbitrary compact carrier K is useful internally, but it leaves the user responsible for choosing a carrier that will later interact well with charts, partitions, and support control. The first-principles theorem fixes the canonical choice and proves the required inclusion automatically. In this way the compactness hypothesis is stated exactly as compact support of the form.

7.2 Pointwise chart boxes from the half-space model

The intrinsic route requires pointwise chart-box data on K . The first-principles theorem does not expose that data. It is generated from the half-space model using self charts:

```
PointwiseCompactSupportChartData.ofHalfSpaceModelSelfCharts
```

The background class

HalfSpaceModel

records that the model range is the upper half-space. From this, the development proves that self-chart coordinates land in the half-space and can be separated into boundary and interior cases.

At an interior coordinate point, one chooses a box contained in the positive normal region. At a boundary coordinate point, one chooses a half-space box whose lower normal coordinate is zero. The constructor proves the neighborhood and box-containment facts required by the selected-cover layer. Thus the pointwise geometry is no longer a user-facing certificate.

7.3 Open shrinks

Closed or half-closed chart boxes are not the right objects for constructing a partition of unity. The formal proof therefore chooses auxiliary open shrinks inside the pointwise chart boxes. This is encoded in `PointwiseCompactSupportChartData.OpenShrink` and:

```
PointwiseCompactSupportChartData.selectedOpenCoverOfPointwise
```

The selected open cover remembers both the open set used for partition construction and the assigned closed or half-space chart box used later for support containment.

The proof pattern is:

pointwise chart-box neighborhood $\Rightarrow$ open shrink inside the chart box $\Rightarrow$ finite
selected open cover by compactness.

This is one of the places where Lean forces a useful distinction. A neighborhood fact is not itself a globally selected open cover with finite index type and assigned box data. The formalization constructs that object.

The full construction chain is:

$$\begin{array}{c}
 \text{pointwise chart-box neighborhood at each } x \in K \\
 \Downarrow \\
 \text{open shrink } U_x \text{ with } x \in U_x \subseteq \text{assigned box} \\
 \Downarrow \\
 \text{finite selected open cover by compactness} \\
 \Downarrow \\
 \text{support-controlled partition subordinate to the selected cover} \\
 \Downarrow \\
 \text{boundary active carriers and finite half-space refinements.}
 \end{array}$$

Every arrow carries data used later. The selected open set is needed for the partition theorem; the assigned box is needed for support containment; the finite index type is needed for endpoint reconstruction.

Lemma 7.1 (Compactness-to-finite-selection). *Pointwise chart-box neighborhood data on a compact carrier K determines a finite selected chart-box cover of K , after choosing open shrinks inside the pointwise boxes. Moreover, the selected cover remembers both the open sets used to build partitions and the assigned coordinate boxes used for later support containment.*

This is mathematically the finite-subcover step, but the formal statement is stronger than a bare finite cover: it preserves enough bookkeeping to connect partition support, chart boxes, and later half-space refinement.

The finite selection has three pieces of information that must travel together. First, it has a finite index type. Second, each index remembers the chosen chart and whether it is treated as an interior or boundary chart. Third, each index remembers an assigned box that contains the support portion of the eventual partition coefficient. Omitting any one of these pieces would force a later theorem to reintroduce a manual compatibility hypothesis.

This is the first major certificate-elimination step in the global proof. Earlier routes accepted selected covers and openness data as public inputs. The final route proves that they are consequences of pointwise chart-box neighborhoods and compactness, and the first-principles route generates the pointwise data from the half-space model.

7.4 Support-controlled partitions

The partition used by the theorem is not merely a family summing to one. It is a support-controlled selected partition:

`OpenSupportControlledSelectedPartition.`

The key guarantee is that the topological support of a partition coefficient, intersected with the compact carrier, lies inside the assigned selected open set and then inside the assigned chart box:

`openSupportControlledSelectedPartition_tsupport_inter_subset_assigned`

This statement is later used twice. First, it proves that the first boundary localization $\rho_i\omega$ is supported in the boundary active carrier. Second, after smooth refinement, it contributes to the proof that each refined localized zero representative is supported inside its half-space support box. Without support control at the partition stage, artificial-face vanishing would have to be assumed later.

The support-controlled partition is stronger than the usual partition of unity existence theorem in exactly the way Stokes needs [GP74, Hir76, War83, Tu11]. It is not enough that $\sum_i \rho_i = 1$ on K . We need

$$\text{tsupp}(\rho_i) \cap K \subseteq U_i$$

for the selected open shrink U_i , and then U_i must itself lie inside the assigned chart box. This is the route by which compactness and partitions eventually become a statement about artificial coordinate faces.

7.5 Boundary active carriers

For a selected boundary index i , the boundary active carrier is the part of K where the boundary partition coefficient is active. Its coordinate image is the set that must be covered by half-space boxes. In the intrinsic route, the selected boundary boxes themselves provide the coordinate ambient sets:

`SupportControlledSelectedPartition.finiteHalfSpaceCoverOfSelectedBoundaryBoxes`

This avoids the older public fields for boundary ambient sets and collar-prism certificates. The theorem now derives the finite half-space cover from the selected boundary box geometry generated earlier.

7.6 Smooth refinement

The finite half-space cover still has to be converted into smooth refined coefficients. This is handled by

```
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedSmoothRefinement
```

The surrounding API proves:

```
selectedSmoothRefinementAmbientOpen
isOpen_selectedSmoothRefinementAmbientOpen
boundaryActiveCarrier_subset_iUnion_selectedSmoothRefinementAmbientOpen
selectedSmoothRefinementAmbientOpen_subset_boundaryChartBox
```

The smooth refinement introduces a sigma-indexed family of pieces. A boundary index i may have many active refined boxes q . The theorem

`selectedSmoothRefinement_boundaryPieces`

connects the refinement's boundary pieces with the selected finite half-space cover. This connection is used by the endpoint route to ensure that the same pieces are used for support, local Stokes, and finite-sum reconstruction.

The refined coefficient has the form

$$\rho_{i,q}(x) = \rho_i(x) \psi_{i,q}(x),$$

where ρ_i is the selected boundary partition coefficient and $\psi_{i,q}$ is the smooth subpartition coefficient subordinate to the refined half-space box. The identity

$$\sum_{q \in Q_i} \rho_{i,q}(x) = \rho_i(x)$$

on the relevant carrier is the algebraic reason the refined local sums reconstruct the original boundary-localized contribution.

The smooth refinement is also where the endpoint index set becomes visible. Before refinement there is one boundary coefficient ρ_i for each selected boundary chart. After refinement the local half-space theorem is applied to the family $\rho_{i,q}$, indexed by a boundary chart and a selected refined box. The endpoint layer must use precisely this same sigma-indexed family; otherwise local Stokes would be proved for one finite sum and the represented endpoint would be defined over another.

7.7 Why finite data is not bookkeeping

The finite selected data is the skeleton of the represented theorem. It determines the index types of the endpoint sums, the coefficients of the localized forms, the boxes on which local Stokes is applied, and the support facts used to remove artificial faces. A mismatch between any two of these finite structures would make the final reconstruction theorem false or unusable.

The first-principles theorem hides the finite data from users because Lean constructs it internally. It does not make the data irrelevant. The data is the mechanism by which compact support becomes a finite coordinate Stokes proof.

In this sense, compactness is not a side condition of Stokes but a generator of the proof route. It selects the charts, fixes the partition coefficients, determines which boundary carriers are active, and supplies the index sets for the represented endpoints. The final theorem can have a small input because this generated finite data is coherent by construction.

Remark 7.2. *The support-controlled partition is the reason compactness can be used once, near the beginning of the proof, rather than reintroduced as local hypotheses for every box. Once the finite selected data is built, all later support facts are consequences of the selected construction.*

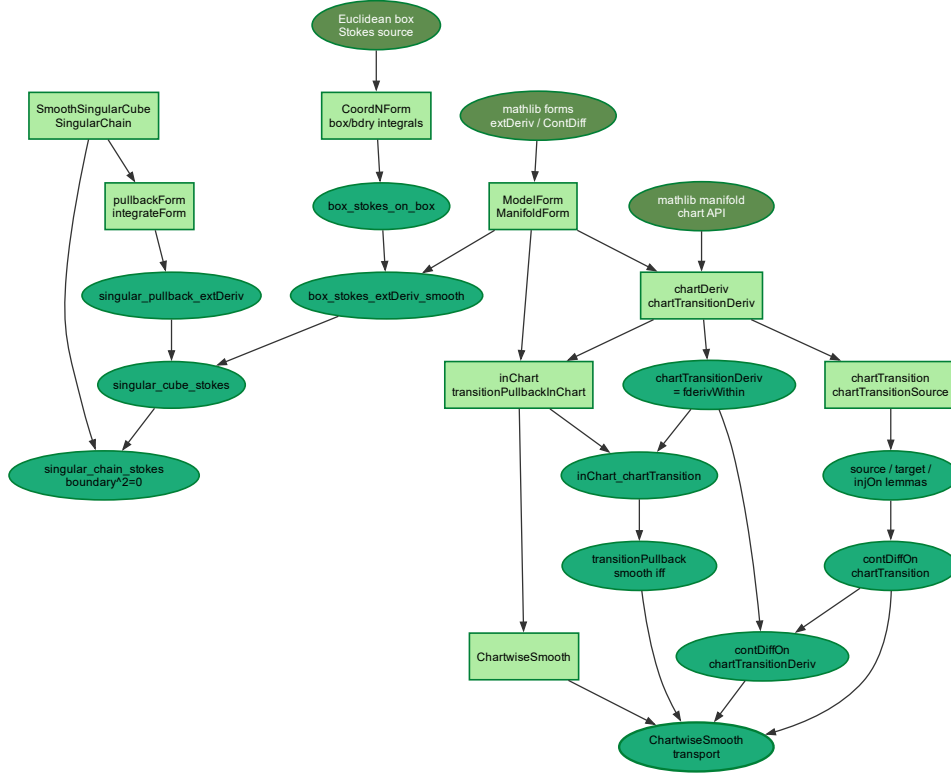


Figure 3: The compactness and refinement core: compact support gives a finite selected cover; support-controlled partitions and smooth refinements prepare the local half-space Stokes data. Light-green rectangular nodes denote definitions or constructed data; dark-green oval nodes denote theorem-level statements or derived equalities.

8 Endpoint Reconstruction

After each refined local piece satisfies half-space Stokes, the proof still has to reconstruct a global represented equality. This section explains why the endpoint layer is a mathematical component of the formalization rather than a renaming of local terms.

8.1 Nested finite sums

The selected boundary refinement is indexed in two stages. There is a finite set of boundary chart indices i , and for each i a finite set of active refined pieces q . The local theorem gives

$$\text{localBulkTerm}(i, q) = \text{localBoundaryTerm}(i, q)$$

for every active pair. Summing gives

$$\sum_i \sum_{q \in Q_i} \text{localBulkTerm}(i, q) = \sum_i \sum_{q \in Q_i} \text{localBoundaryTerm}(i, q).$$

In Lean, this is not merely an algebraic observation. The finite sets Q_i must be definitionally or propositionally connected with the selected half-space cover, the smooth refinement, and the local Stokes data. The endpoint structure is designed to keep those indices aligned.

The guiding invariant is that the same active pair (i, q) performs three roles at once: it chooses the half-space box, chooses the refined coefficient, and appears in the endpoint sum. If any of these are separated, the proof acquires adapter fields whose only purpose is to prove that two finite descriptions are the same. The localized endpoint layer avoids this by generating the endpoint from the refined pieces that already carry the local Stokes data.

Lemma 8.1 (Finite endpoint summation). *Given a localized refined endpoint with finite boundary pieces Q_i , if*

$$B_{i,q} = C_{i,q}$$

for every active refined piece $q \in Q_i$, then the generated represented endpoints satisfy

$$\sum_i \sum_{q \in Q_i} B_{i,q} = \sum_i \sum_{q \in Q_i} C_{i,q}.$$

The Lean endpoint layer instantiates $B_{i,q}$ as `localBulkTerm i q` and $C_{i,q}$ as `localBoundaryTerm i q`.

The lemma looks elementary, but its Lean version is valuable because it fixes the finite index family once and for all. The same membership proof $q \in Q_i$ is used to access the box corners, the coefficient, the local form, the zero-support theorem, and the local Stokes fields. This prevents the proof from silently changing finite sums when moving between local and global layers.

The proof itself has three algebraic steps. First, it rewrites the generated bulk endpoint as the nested sum of the local bulk terms. Second, it rewrites each local bulk term using the per-box Stokes equality. Third, it folds the resulting nested sum back into the generated boundary endpoint. In a paper proof these are often collapsed into the phrase “summing over the cover.” In Lean they are separate because every sum carries its index type and membership conditions.

8.2 Localized refined partitions

The core endpoint structure is

`CoverIndexedBoundaryLocalizedRefinedPartition.`

It stores the refined boundary pieces and the generated local bulk and boundary terms. Its definitions

`generatedRepresentedBulkIntegral`

and

`generatedRepresentedBoundaryIntegral`

are the represented endpoints at this layer. The theorem

`representedStokes_of_localStokes`

turns a finite family of local equalities into equality of these generated endpoints.

The zero-localized route refines this through

`CoverIndexedBoundaryLocalizedRefinedPartition.ZeroLocalStokesData.`

This record stores the specific local data needed to apply the half-space local theorem to zero-localized representatives. It includes source and target charts, local forms, box corners, zero-support information, and the local terms whose equality will be summed.

The localized refined partition is intentionally smaller than the older global endpoint certificates. It does not attempt to encode every possible future native integral comparison. It records the data needed to state and prove the represented finite-sum equality. This makes it a stable target: future comparison theorems can map from this generated endpoint to native integrals without changing the local Stokes proof.

This design also prevents the represented theorem from depending on a future choice of native manifold integration API. The endpoint structure fixes the finite coordinate equality now; a later native bridge can prove that this finite coordinate quantity agrees with a chosen global integral definition.

8.3 From zero local Stokes data to endpoints

The theorem

`ZeroLocalStokesData.localStokes_of_interiorFields`

builds the per-box local Stokes equality from interior fields and zero support. Then

`boundaryHalfSpaceBulkSum_eq_trueBoundarySum_of_zeroLocalStokesData`

proves the nested finite-sum equality. Finally,

`representedStokes_of_zeroLocalStokesData`

reconstructs equality of the generated represented endpoints.

Theorem 8.2 (Zero-localized endpoint reconstruction). *For the localized refined partition generated by the intrinsic route, the combination of zero support and half-space interior fields implies equality of the generated represented bulk and boundary endpoints. In Lean this is the role of:*

CoverIndexedBoundaryLocalizedRefinedPartition.
`representedStokes_of_zeroLocalStokesData`

The first-principles route uses the constructor

`hasIntrinsicRoute_of_selectedSmoothRefinement`

to assemble the endpoint, zero-localized local data, boundary piece equality, zero support, and interior fields into an intrinsic certificate. Once the canonical selected smooth refinement supplies the last two ingredients, the intrinsic route has a canonical endpoint equality.

The role of this constructor is precisely to avoid exposing an endpoint adapter to the final user. It proves that the endpoint, local Stokes data, zero-support theorem, and regularity fields are all generated from the same selected smooth refinement. Thus the endpoint equality is not imported from outside the proof; it is assembled from the local Stokes theorem over the canonical finite route.

8.4 Canonical endpoints at the top level

The intrinsic input defines

`canonicalRepresentedBulkIntegral` and `canonicalRepresentedBoundaryIntegral`

by choosing the route generated from the canonical selected smooth refinement:

`hasIntrinsicRoute_intrinsic.`

The first-principles input then projects these endpoints through its generated intrinsic input. Thus the top-level definitions are not external names. They are the endpoint definitions induced by the canonical finite selected route.

8.5 Why reconstruction matters

Endpoint reconstruction is where the represented theorem becomes global. The half-space local theorem knows only about one coordinate box. Zero-localized support knows only how to kill artificial faces for one refined representative. Compactness and refinement know how to generate finitely many such pieces. The endpoint layer proves that all of these local facts assemble into one equality between the two canonical represented quantities associated with the input form.

This is also the layer that makes the represented-to-native future work well-posed. A future theorem comparing represented endpoints with native manifold integrals can target

`canonicalRepresentedBulkIntegral` and `canonicalRepresentedBoundaryIntegral`

directly. It does not need to rediscover the local proof route.

There is an important conceptual payoff here. The proof does not define a global endpoint first and then ask local lemmas to match it. Instead, the endpoint is generated from the same local pieces used by the proof. This removes a common source of mismatch in large formalizations: the local theorem, the summation theorem, and the final theorem all quantify over the same finite piece data.

This is why the word canonical is warranted in the represented theorem. It does not mean independent of every possible choice of partition or chart cover; that would be the native comparison problem. It means canonical relative to the generated first-principles route: once the input form is fixed, the route constructs a selected finite proof object, and the endpoints are the finite sums determined by that object.

8.6 What remains for native integrals

Endpoint reconstruction stops at the finite coordinate equality. To obtain a future native theorem, one must prove two comparison results. The bulk comparison identifies the represented bulk sum with the chosen definition of $\int_M d\omega$. The boundary comparison identifies the represented lower-face sum with the chosen definition of $\int_{\partial M} \omega$ under the global boundary orientation. These comparisons are logically downstream from the represented theorem: they should not be mixed into the proof that the finite coordinate endpoints are equal.

9 Main Lean Theorem

The preferred public theorem is located in

`Stokes/Global/CoverIndexedZeroCompactRepresentedStokesFirstPrinciples.lean`.

It is re-exported through the slim public cover-indexed import. This section records the theorem shape and the internal route.

9.1 Ambient Lean context

The theorem is stated for a space M charted over a model H , with model with corners

`I : ModelWithCorners Real (Fin (n + 1) -> Real) H.`

The surrounding Lean context includes the usual manifold and topological assumptions used by the route, expressed using the typeclass and structure mechanisms of Lean and mathlib [dMKA⁺15, dMU21, Baa22, The20]:

`IsManifold I top M, T2Space M, SigmaCompactSpace M,`

finite dimensionality of the coordinate model, and the half-space model assumption

`HalfSpaceModel I.`

The half-space model is a background geometric assumption: it tells Lean that the model range is the upper half-space and supports the automatic construction of pointwise boundary and interior chart boxes.

9.2 First-principles input

The public input structure is:

```
structure CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput
  (I : ModelWithCorners Real (Fin (n + 1) -> Real) H)
  (omega : ManifoldForm I M n) where
  chartwiseSmooth : ManifoldForm.ChartwiseSmooth I omega
  compactSupport : IsCompact (closure (ManifoldForm.support I omega))
```

There is no public selected cover, no partition field, no boundary ambient field, no collar-prism certificate, no local Stokes field, and no endpoint adapter. These objects are generated internally.

The compact carrier is:

```
def supportCarrier : Set M :=
  closure (ManifoldForm.support I omega)
```

The intrinsic input is built by:

```
def intrinsicInput :
  CoverIndexedZeroCompactRepresentedStokesIntrinsicInput
  (I := I) (omega := omega) supportCarrier where
  pointwise :=
    PointwiseCompactSupportChartBoxData.ofHalfSpaceModelSelfCharts
    (I := I) supportCarrier
  hK := X.compactSupport
  chartwiseSmooth := X.chartwiseSmooth
  support_subset_K := subset_closure
```

This is the first major compression: the four-field intrinsic theorem is generated from the two-field first-principles theorem.

9.3 Canonical represented endpoints

The top-level endpoints are:

```
def canonicalRepresentedBulkIntegral : Real :=
  (X.intrinsicInput (I := I) (omega := omega)).
  canonicalRepresentedBulkIntegral

def canonicalRepresentedBoundaryIntegral : Real :=
  (X.intrinsicInput (I := I) (omega := omega)).
  canonicalRepresentedBoundaryIntegral
```

They inherit their meaning from the intrinsic route: finite selected cover, selected smooth refinement, zero-localized local data, half-space local Stokes, and endpoint reconstruction.

9.4 The theorem

The theorem is:

```
theorem representedStokes :
  X.canonicalRepresentedBulkIntegral (I := I) (omega := omega) =
    X.canonicalRepresentedBoundaryIntegral (I := I) (omega := omega) := by
  simp [canonicalRepresentedBulkIntegral,
        canonicalRepresentedBoundaryIntegral] using
    (X.intrinsicInput (I := I) (omega := omega)).
    representedStokes
```

The proof is short because the intrinsic theorem has already generated the route:

```
selectedSmoothRefinement
zeroSupportOfSelectedSmoothRefinement
interiorFieldsOfSelectedSmoothRefinement
hasIntrinsicRoute_intrinsic
representedStokes
```

9.5 Intrinsic route

The intrinsic structure still has four mathematical fields:

```
structure CoverIndexedZeroCompactRepresentedStokesIntrinsicInput
  (K : Set M) where
  pointwise : PointwiseCompactSupportChartBoxData I K
  hK : IsCompact K
  chartwiseSmooth : ManifoldForm.ChartwiseSmooth I omega
  support_subset_K : ManifoldForm.support I omega <= K
```

It is useful as an internal theorem-facing layer. It states the theorem with a chosen compact carrier and pointwise chart-box data. The first-principles theorem hides both by choosing $K = \text{supp}(\omega)$ and generating pointwise data from the half-space model.

The intrinsic route defines the selected cover, partition, and boundary finite cover through:

```
openSelectedCover
selectedCover
openSelectedPartition
selectedPartition
selectedBoundaryFiniteCover
```

The canonical smooth refinement is generated by

selectedSmoothRefinement.

The two critical automatically generated proof obligations are:

zeroSupportOfSelectedSmoothRefinement

and

interiorFieldsOfSelectedSmoothRefinement.

Together they feed into

hasIntrinsicRoute_intrinsic.

Theorem 9.1 (First-principles represented Stokes). *In the half-space model context, chartwise smoothness of ω and compactness of $\text{supp}(\omega)$ imply equality of the canonical represented bulk and boundary endpoints generated by the compact-support local-to-global route.*

The proof of [Theorem 9.1](#) is not an invocation of a black-box global theorem. It is the composition of the following generated statements: compactness produces selected finite data; selected finite data produces a smooth half-space refinement; the refinement produces zero-localized support and analytic interior fields; local half-space Stokes gives each refined identity; and endpoint reconstruction sums those identities.

9.6 Reading the theorem mathematically

Mathematically, the final theorem says:

For a chartwise smooth n -form on a half-space-modeled smooth manifold, if the closure of its support is compact, then the finite coordinate bulk endpoint generated by the compact-support Stokes proof equals the finite coordinate boundary endpoint generated by the same proof.

This is the clean statement the development was organized to reach. The large amount of internal code exists so that the theorem can be stated in this small form without making the user provide the hidden proof objects.

10 Artifact and Scale

The project is a large Lean artifact, not a short isolated formalization. The active `Stokes/` directory contains 544 Lean files and approximately 165006 Lean lines. This scale reflects both the mathematics and the proof engineering required to compress a certificate-heavy development into a small public theorem. The maintenance and interface issues are in the same family as those discussed for large mathematical libraries such as mathlib [[vDEL20](#), [Baa22](#), [BBC⁺25](#)].

10.1 Repository organization

The artifact repository is available at <https://github.com/evaristebernhard/Stokes> (accessed 29 May 2026). The active Lean files live under `Stokes/`. The main public route is exported by

`Stokes.Global.CoverIndexedFirstPrinciples`

and by the slimmed cover-indexed public entry

`Stokes.Global.CoverIndexed.`

Historical routes remain available as implementation or compatibility modules, but the default public surface is centered on the first-principles theorem.

The proof archaeology documents used to prepare this paper live under `docs/formalization-log/`. They record the mathematical bottlenecks behind support, zero-localization, half-space boundaries, compact finite selection, endpoint reconstruction, and API slimming. These documents are not part of the trusted proof base, but they are useful for explaining the artifact.

10.2 Public API slimming

During development, several intermediate theorem shapes exposed internal certificates: raw selected covers, clean inputs, natural inputs, collar routes, endpoint adapters, image-control packages, local Stokes fields, and manual support assumptions. These stages were useful

because they stabilized the statement while independent local obligations were being proved. The final API is the result of removing those theorem-facing fields.

The first-principles theorem is therefore best viewed as a compression result, but not in the superficial sense of putting a small facade around a large record. It proves that the records which earlier routes consumed are generated by the mathematical hypotheses. In this sense the artifact moves from a certificate-consuming interface to a certificate-generating interface. It does not bypass the hidden proof objects. It constructs them internally and then applies the intrinsic represented theorem. This explains why a short top-level proof can rest on a large body of Lean code.

10.3 Route evolution

The development history is visible in route names such as Raw, Clean, Natural, Intrinsic, and FirstPrinciples. These names should not be read as competing theorems of equal status. They record a field-elimination process.

Route style	Main theorem-facing burden
Raw	selected cover and low-level local data are explicit
Clean	several geometric certificates are packaged but still visible
Natural	generated data starts replacing manual endpoint fields
Intrinsic	pointwise chart-box data, compactness, and support control remain
FirstPrinciples	only chartwise smoothness and compact closed support remain

The engineering point is that intermediate routes made hidden obligations visible. Once an obligation could be generated from earlier mathematical data, it moved out of the public input. The final theorem is therefore not another layer of ceremony; it is the accumulated result of proving that the former certificate fields are consequences of the natural hypotheses.

10.4 Verification gates

The development is checked by the standard Lean build commands. For the paper claim, the relevant commands are:

```
lake build Stokes
rg "\bsorry\b|\badmit\b|\^s*axiom\b" --glob "*.lean"
```

The theorem-level axiom audit is:

```
#print axioms
Stokes.CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.
  representedStokes
```

The expected output contains the standard Lean/mathlib axioms:

`propext, Classical.choice, Quot.sound.`

This paper rewrite does not rerun the full Lean build, because it modifies only `paper/`. The Lean names quoted in the text were checked by lightweight repository search against the active sources.

10.5 What scale bought

The artifact's size bought several mathematical reductions:

- ordinary total chart support assumptions were replaced by zero-localized support theorems;

- boundary-chart ambient-open assumptions were replaced by half-space interior and closed-box regularity fields;
- user-supplied finite covers and refinements were replaced by compactness-driven selected constructions;
- endpoint adapters were replaced by generated localized refined endpoints;
- theorem-facing geometric certificates were internalized behind the first-principles route.

These reductions are the main proof-engineering achievement of the project. They also make the theorem usable as a stable milestone: future work can build native integral comparison theorems against the canonical represented endpoints rather than against a moving collection of intermediate certificates.

The zero-localized support reduction is representative of the whole pattern. The old theorem-facing hypothesis would have asked the user to control the support of an ordinary totalized chart representative. The final route proves instead that the generated zero-localized representative has the required topological support. This removes a bad hypothesis by replacing the object of the theorem and then proving the comparison and support facts needed to use it.

10.6 Artifact reading guide

A reader who wants to audit the mathematics should start from the first-principles theorem, then follow the intrinsic route backward: canonical endpoint, intrinsic route certificate, selected smooth refinement, zero support, interior fields, and half-space local Stokes. A reader who wants to audit the engineering should instead compare the historical route modules with the public import surface. The key question is not how many modules exist, but which assumptions remain theorem-facing. In the current preferred API, the answer is exactly the first-principles pair: chartwise smoothness and compact closed support.

11 Related Work

11.1 Lean and mathlib

The formalization is implemented in Lean 4 [dMKA⁺15, dMU21, Lea26, AdMKU26] and builds on mathlib [The20, The26, vDEL20, BBC⁺25]. The project follows the mathlib-compatible strategy of using existing topology, analysis, manifolds, finite-dimensional calculus, and exterior-algebra infrastructure where possible. The local box Stokes layer is organized around mathlib-style divergence and differentiability interfaces rather than a separate analytic foundation.

The theorem also exposes places where future mathlib APIs would be valuable. In particular, the represented-to-native bridge would naturally depend on a stable API for integrating differential forms on oriented manifolds with boundary, together with chart-independence and boundary-orientation comparison theorems.

11.2 Prior Lean Stokes artifacts

The repository treats `d0d1/lean-stokes-theorem` as prior work and a Lake dependency reference [d0d26]. That project is useful prior art for Stokes formalization, especially around cubical or smooth-singular-cube approaches. Because of licensing considerations, this project

uses external developments as references rather than as copied source. The present contribution is different in emphasis: it formalizes a compact-support coordinate-represented local-to-global route through chart boxes, support-controlled partitions, half-space boundary boxes, zero-localized representatives, and generated represented endpoints.

11.3 Classical manifold treatments

Classical treatments of manifolds and differential forms provide the mathematical background for chartwise local arguments, partitions of unity, compact support, boundary orientations, and Stokes’ theorem [Spi65, Mun91, GP74, BT82, Hir76, War83, AMR88, Lan95, Tu11, Lee13, Mor01]. The formalization can be read as a detailed elaboration of the proof steps that such texts compress. The point is not that those steps are mathematically surprising to a geometer. The point is that their exact dependencies become visible when the proof is checked by Lean.

This is especially visible for boundary charts. Classical accounts freely move between the local half-space model, the true boundary face, and the auxiliary faces of a coordinate box. In Lean these become separate theorems: the lower face must be identified with the boundary contribution, the sign must match the outward-first convention, and every artificial face must be removed by a support theorem.

11.4 Formalized mathematics and proof engineering

The work also sits in a broader theorem-proving ecosystem. Coq/Rocq, Isabelle/HOL, and HOL Light have long provided reference points for formalized mathematics and proof engineering [BC04, NPW02, Har09, Wie06, HUW14]. Within that ecosystem, formalized analysis and geometry have repeatedly required substantial library and interface work, for example filters and analysis in Isabelle/HOL, smooth manifolds in Isabelle/HOL, formalized Haar measure, and the change-of-variables theorem in mathlib [HH13, IZ19, vD21, Gou22].

Large formalizations often spend much of their effort on interfaces: finding the right theorem boundary, replacing manual certificates by constructors, and organizing local lemmas so that global statements have natural hypotheses. Prominent formalization projects such as CompCert [Ler09], the formal four-color theorem [Gon08], the machine-checked odd-order theorem [GAA⁺13], the formal proof of the Kepler conjecture [HAB⁺17], Schemes in Lean [BCM22], formalized Galois theory [BL22], and the Liquid Tensor Experiment [CMS22] illustrate that the architecture of a large proof artifact is part of the contribution. This project is an example in differential geometry and analysis. The intermediate routes were not abandoned failures; they were stages in identifying which fields were true mathematical hypotheses and which were internal proof objects. The first-principles theorem is the compressed result of that interface work.

This distinction matters for evaluating formalized mathematics. A certificate-consuming theorem may be enough to close a proof internally, but it does not yet present the theorem in the form a mathematician expects to use. The present artifact treats assumption compression as part of the mathematical work. Covers, refinements, support facts, local regularity records, and endpoint reconstruction are not left as user obligations once they can be generated from compact support and chartwise smoothness. This agrees with a broader view of formalization as both theorem proving and mathematical interface design [Avi18, Baa22].

The support and zero-localization layer is also an example of a common phenomenon in formalization: a statement that is harmless on paper becomes false or ill-shaped when all functions are total. The solution is not merely a technical patch. It identifies the correct formal representative of the informal mathematical object.

12 Limitations and Future Work

The theorem proved by the artifact is a coordinate-represented compact-support Stokes theorem. This is a strong local-to-global result, but it is not the final possible theorem about native manifold integrals. This section states the boundary of the claim and the natural next steps.

12.1 Represented-to-native comparison

The main remaining mathematical bridge is to identify the generated represented endpoints with a chosen native integration API. The desired comparison would identify `canonicalRepresentedBulkIntegral` with $\int_M d\omega$ and `canonicalRepresentedBoundaryIntegral` with $\int_{\partial M} \omega$. Such a bridge should prove that the finite coordinate sums generated by the local proof are independent of the selected cover and partition and agree with the native integral definitions. This requires a stable target API for integration of differential forms on oriented smooth manifolds with boundary.

The current theorem deliberately stops before this comparison. It proves the finite coordinate equality generated by Stokes, leaving the comparison with native integrals as a separate theorem layer.

12.2 Orientation and boundary APIs

The half-space layer already proves the local sign bridge between the lower normal face and the outward-first convention. A full native theorem will need to connect this local sign convention to a global boundary orientation API. That includes transporting orientations through charts, proving compatibility under chart changes, and connecting boundary measures or integrals with the coordinate lower-face terms.

12.3 Consolidation

The repository still contains historical routes and compatibility modules. They are valuable as development records and as local theorem sources, but the public API should remain centered on the first-principles route. Future engineering work can continue to consolidate implementation modules, move historical theorem shapes out of default imports, and keep theorem-facing names focused on the final mathematical interface.

12.4 Possible extensions

Several extensions are natural:

- comparison of represented endpoints with native integrals;
- a compact oriented smooth manifold with boundary theorem once the native integration bridge is in place;
- more systematic chart-independence lemmas for represented endpoints;
- upstreaming general-purpose half-space and support-localization lemmas to `mathlib` where appropriate;
- a shorter conference version emphasizing the artifact architecture and verification gates.

These are extensions of the current milestone, not repairs of it. The present artifact already proves the generated coordinate Stokes equality from first-principles compact-support hypotheses.

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A Key Lean Statements

This appendix records simplified versions of the main Lean statements. The snippets omit universe parameters, typeclass context, and many implicit arguments. They are intended as a reader's guide to the formal theorem shapes, not as copy-paste source.

A.1 First-principles input and theorem

```
structure CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput
  (I : ModelWithCorners Real (Fin (n + 1) -> Real) H)
  (omega : ManifoldForm I M n) where
  chartwiseSmooth : ManifoldForm.ChartwiseSmooth I omega
  compactSupport : IsCompact (closure (ManifoldForm.support I omega))

def supportCarrier : Set M :=
  closure (ManifoldForm.support I omega)

def canonicalRepresentedBulkIntegral : Real := ...
def canonicalRepresentedBoundaryIntegral : Real := ...

theorem representedStokes :
  X.canonicalRepresentedBulkIntegral =
  X.canonicalRepresentedBoundaryIntegral
```

Mathematically, this says that chartwise smoothness and compactness of the closed support generate the represented local-to-global Stokes route and prove equality of the generated endpoints.

A.2 Zero-localized chart representative

```
def ManifoldForm.transitionPullbackInChartZero
  (x0 x1 : M) (omega : ManifoldForm I M n) :
  (Fin (n + 1) -> Real) ->
  (Fin (n + 1) -> Real) [/\^Fin n]->L[Real] Real := ...

theorem transitionPullbackInChartZero_support_subset_source :
  Function.support (transitionPullbackInChartZero I x0 x1 omega) <=
  (extChartAt I x0).target

theorem transitionPullbackInChartZero_eqOn_transitionPullbackInChart :
  EqOn (transitionPullbackInChartZero I x0 x1 omega)
  (transitionPullbackInChart I x0 x1 omega)
  (ManifoldForm.chartTransitionSource I x0 x1)
```

The zero representative is the formal substitute for the informal convention that chart representatives are considered only where charts are valid.

A.3 Selected zero support

```

theorem zeroSupportOfSelectedSmoothRefinement :
  forall i (q : Fin (n + 1) -> Real),
    q in selectedBoundaryFiniteCover.activePieces i ->
      tsupport
        (ManifoldForm.transitionPullbackInChartZero I
          (selectedCover.boundaryChart i.1)
          (selectedCover.boundaryChart i.1)
          (ManifoldForm.localizedForm I
            (selectedSmoothRefinement.coefficient i <i,q>) omega))
    <= halfSpaceSupportBox
      (selectedBoundaryFiniteCover.lowerCorner i q)
      (selectedBoundaryFiniteCover.upperCorner i q)

```

This is the theorem that turns support-controlled partitions and smooth refinement into the support condition used by half-space local Stokes.

A.4 Half-space local Stokes

```

structure HalfSpaceBoxInteriorStokesFields
  (omega : CoordNForm n) (a b : Fin (n + 1) -> Real) where
  le : a <= b
  lower_zero : a 0 = 0
  continuous_signedCoeff : ...
  hasFDerivAt_signedCoeff : ...
  integrable_divergence : ...
  bulk_eq_boxIntegral : ...

theorem halfSpaceLocalStokes_compactSupport_of_interiorFields
  (D : HalfSpaceBoxInteriorStokesFields omega a b)
  (hsupp : boxFaceCoeffTSupportInHalfSpaceBox omega a b) :
  halfSpaceLocalBulkIntegral omega a b =
    halfSpaceBoundarySign n * halfSpaceBoundaryFormIntegral omega a b

```

The local theorem isolates the true boundary face and removes all artificial faces through the support condition.

A.5 Automatic interior fields

```

def interiorFieldsOfSelectedSmoothRefinement :
  forall i, i in Finset.univ ->
    forall q, q in selectedBoundaryFiniteCover.activePieces i ->
      HalfSpaceBoxInteriorStokesFields
        (ManifoldForm.transitionPullbackInChart I
          (selectedCover.boundaryChart i.1)
          (selectedCover.boundaryChart i.1)
          (ManifoldForm.localizedForm I
            (selectedSmoothRefinement.coefficient i <i,q>) omega))
        (selectedBoundaryFiniteCover.lowerCorner i q)
        (selectedBoundaryFiniteCover.upperCorner i q)

```

This constructor is where chartwise smoothness of ω , smoothness of the refined coefficient, and selected box geometry produce the local analytic data needed by the half-space theorem.

A.6 Endpoint reconstruction

```

def CoverIndexedBoundaryLocalizedRefinedPartition.
  generatedRepresentedBulkIntegral : Real := ...

def CoverIndexedBoundaryLocalizedRefinedPartition.
  generatedRepresentedBoundaryIntegral : Real := ...

theorem representedStokes_of_zeroLocalStokesData :

```

<pre>endpoint.generatedRepresentedBulkIntegral = endpoint.generatedRepresentedBoundaryIntegral</pre>
--

The endpoint theorem sums the per-box local Stokes equalities over the finite boundary chart and refined piece indices, producing the represented global equality.

B Proof Obligations and Discharge Points

This appendix gives a more operational map of the proof. Each item records a mathematical obligation that appears in the compact-support Stokes proof and the layer where the Lean development discharges it. The list is useful because many early theorem statements exposed these obligations as input fields; the first-principles theorem hides them by proving the corresponding constructors.

B.1 Carrier and support obligations

Compact carrier. The proof needs a compact set K containing the support of the form. The first-principles theorem chooses

$$K = \overline{\text{supp}(\omega)}.$$

The public hypothesis is compactness of this set. The inclusion $\text{supp}(\omega) \subseteq K$ is discharged by closure monotonicity, not by a user-provided field.

Localized support. For each partition coefficient ρ_i , the support of $\rho_i\omega$ must lie in the corresponding active carrier. This is discharged by combining two facts: a localized form can be nonzero only where the coefficient is nonzero and where the original form is nonzero; the latter points lie in the compact carrier. This obligation is used again after refinement, when $\rho_{i,q}\omega$ is rewritten as a second localization of $\rho_i\omega$.

Topological support. Local Stokes does not merely need ordinary support. It needs a topological support inclusion strong enough to prove artificial face integrals vanish. The proof therefore upgrades ordinary support control to `tsupp` control using closedness of the selected half-space support boxes.

B.2 Cover and partition obligations

Pointwise chart boxes. For each point of the compact carrier, the proof needs a chart box adapted to the half-space model. The first-principles route generates this data from self charts and the `HalfSpaceModel` assumption. Boundary points receive boxes whose lower normal coordinate is zero; interior points receive boxes away from the boundary face.

Open shrink. The assigned chart boxes are not themselves the right objects for partition construction. The proof chooses open shrinks inside them. This produces an open cover suitable for a partition of unity while preserving the assigned box needed later for support.

Finite selected cover. Compactness extracts finitely many of the open shrinks. The selected cover remembers the finite index, the selected chart, the open set, and the assigned box. This is more structured than an ordinary finite subcover because later theorems need all four pieces of data.

Support-controlled partition. The partition must satisfy a support theorem, not only a sum-to-one theorem. For each selected index j , the development proves

$$\text{tsupp}(\rho_j) \cap K \subseteq U_j,$$

where U_j is the selected open shrink. Since U_j is contained in the assigned chart box, this becomes the first support input for the local Stokes route.

B.3 Boundary refinement obligations

Boundary active carrier. Only the boundary-selected partition coefficients contribute to the boundary route. For each boundary index i , the active carrier records the part of K where the boundary coefficient can matter. Its coordinate image is the set to be covered by half-space boxes.

Finite half-space cover. The active coordinate carrier is covered by finitely many half-space boxes. In the final route, these boxes are generated from the selected boundary box data rather than supplied as a public ambient/collar certificate.

Smooth refinement. The finite half-space cover is converted into smooth subpartition coefficients $\psi_{i,q}$. The refined coefficient is

$$\rho_{i,q} = \rho_i \psi_{i,q}.$$

The refinement must prove both algebraic reconstruction $\sum_q \rho_{i,q} = \rho_i$ on the active carrier and support containment of each $\psi_{i,q}$ in its assigned refined box.

B.4 Local Stokes obligations

Zero representative. The local theorem is applied to a coordinate representative. Because ordinary chart representatives are total functions, the proof uses the zero-localized representative. The obligation is twofold: it must agree with the ordinary representative on the chart source, and it must be supported in the assigned box globally.

Artificial-face support. For each refined local form, the proof must show that the topological support of its zero representative lies in the half-space support box. This is the role of `zeroSupportOfSelectedSmoothRefinement`. The local half-space theorem then converts this support inclusion into vanishing of all artificial faces.

Interior fields. For each refined box, the proof must supply the analytic fields used by box Stokes: closed-box continuity of signed coefficients, differentiability on the open interior, integrability of the divergence term, and the bridge between the exterior derivative coefficient and coordinate divergence. The selected regularity constructor proves these fields from chartwise smoothness and smoothness of the refined coefficient.

Boundary sign. After artificial faces vanish, the remaining lower normal face must be identified with the boundary contribution. The half-space boundary layer proves that the lower face carries the outward-first sign. This is the local orientation content currently needed by the represented theorem.

B.5 Endpoint obligations

Per-box equality. The half-space theorem gives equality of local bulk and true boundary terms for each active refined piece. These are the equalities that will be summed.

Finite-sum reconstruction. The endpoint layer proves that summing the local bulk terms over the selected finite pieces gives the generated represented bulk endpoint, and similarly for the boundary terms. The same finite index family is used throughout, avoiding any hidden reindexing mismatch.

Top-level compression. The intrinsic theorem packages all generated obligations into a route certificate. The first-principles theorem then generates the intrinsic input from chartwise smoothness and compact closed support. Thus the final public theorem has small input not because the obligations disappeared, but because they have named discharge points in the proof.

C Theorem and API Map

This appendix lists the main Lean names used by the first-principles represented Stokes route. It is not a complete module index. The goal is to help readers locate the mathematical layers discussed in the paper.

C.1 Public first-principles route

```
Stokes/Global/CoverIndexedZeroCompactRepresentedStokesFirstPrinciples.lean
Stokes/Global/CoverIndexedFirstPrinciples.lean
Stokes/Global/CoverIndexed.lean
```

```
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.supportCarrier
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.intrinsicInput
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.
  canonicalRepresentedBulkIntegral
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.
  canonicalRepresentedBoundaryIntegral
CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.representedStokes
```

C.2 Half-space model and pointwise chart boxes

```
Stokes/Global/HalfSpaceModel.lean
Stokes/Global/PointwiseChartBoxFromHalfSpaceModel.lean
```

```
HalfSpaceModel
HalfSpaceModel.extChartAt_target_subset_upperHalfSpace
HalfSpaceModel.self_coord_mem_upperHalfSpace
HalfSpaceModel.boundaryChartBoxNeighborhood_mem_nhds_of_coordWithin
PointwiseCompactSupportChartBoxData.ofHalfSpaceModelSelfCharts
```

C.3 Intrinsic selected cover and partition

```
Stokes/Global/CoverIndexedZeroCompactRepresentedStokesIntrinsic.lean
Stokes/Global/CoverIndexedPointwiseOpenSelection.lean
Stokes/Global/CoverIndexedOpenSupportControlledPartition.lean
```

```

CoverIndexedZeroCompactRepresentedStokesIntrinsicInput
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.openSelectedCover
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedCover
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.openSelectedPartition
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedPartition
PointwiseCompactSupportChartBoxData.selectedOpenCoverOfPointwise
PointwiseCompactSupportChartBoxData.selectedOpenSupportControlledPartition
OpenSupportControlledSelectedPartition

```

C.4 Boundary finite cover and smooth refinement

```

Stokes/Global/CoverIndexedSelectedBoundaryBoxFiniteCover.lean
Stokes/Global/CoverIndexedIntrinsicSmoothRefinementConstructor.lean

```

```

SupportControlledSelectedPartition.finiteHalfSpaceCoverOfSelectedBoundaryBoxes
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedBoundaryFiniteCover
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedSmoothRefinementAmbientOpen
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.selectedSmoothRefinement
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.
  selectedSmoothRefinement_boundaryPieces

```

C.5 Zero-localized support

```

Stokes/ManifoldFormZero.lean
Stokes/Global/CoverIndexedZeroLocalizedSupport.lean
Stokes/Global/CoverIndexedIntrinsicZeroSupportConstructor.lean

```

```

zeroOutside
support_zeroOutside_subset
ManifoldForm.transitionPullbackInChartZero
ManifoldForm.transitionPullbackInChartZero_support_subset_source
ManifoldForm.transitionPullbackInChartZero_support_subset_target
ManifoldForm.transitionPullbackInChartZero_support_subset_overlap
ManifoldForm.transitionPullbackInChartZero_eqOn_transitionPullbackInChart
ManifoldForm.transitionPullbackInChartZero_localizedForm
ManifoldForm.transitionPullbackInChartZero_localizedForm_tsupport_subset_halfSpaceSupportBox_of_support_subset_K
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.zeroSupportOfSelectedSmoothRefinement

```

C.6 Selected regularity and half-space local Stokes

```

Stokes/Global/CoverIndexedIntrinsicSelectedRegularityConstructor.lean
Stokes/HalfSpace/BoxInteriorSmoothness.lean
Stokes/HalfSpace/BoxInteriorStokes.lean
Stokes/HalfSpace/Faces.lean
Stokes/HalfSpace/BoundaryIntegral.lean

```

```

CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.
  interiorFieldsOfSelectedSmoothRefinement
HalfSpaceBoxInteriorStokesFields
HalfSpaceBoxOpenInteriorSmoothnessFields
HalfSpaceBoxInteriorStokesFields.ofOpenInteriorSmoothness
halfSpaceLocalStokes_compactSupport_of_openInteriorSmoothness
halfSpaceLocalStokes_compactSupport_of_interiorFields
halfSpaceLocalBoundaryRemainder_eq_zero_of_tsupport_subset_halfSpaceSupportBox
halfSpaceBoundarySign_eq_outwardFirstBoundaryOrientationSign

```

C.7 Localized endpoint and intrinsic route

```
Stokes/Global/CoverIndexedLocalizedRefinedEndpoint.lean
Stokes/Global/CoverIndexedZeroLocalizedIntrinsicCertificate.lean
Stokes/Global/CoverIndexedIntrinsicEndpointConstructor.lean
Stokes/Global/CoverIndexedIntrinsicRouteConstructor.lean
```

```
CoverIndexedBoundaryLocalizedRefinedPartition
CoverIndexedBoundaryLocalizedRefinedPartition.generatedRepresentedBulkIntegral
CoverIndexedBoundaryLocalizedRefinedPartition.generatedRepresentedBoundaryIntegral
CoverIndexedBoundaryLocalizedRefinedPartition.representedStokes_of_localStokes
CoverIndexedBoundaryLocalizedRefinedPartition.ZeroLocalStokesData
CoverIndexedBoundaryLocalizedRefinedPartition.ZeroLocalStokesData.
  localStokes_of_interiorFields
CoverIndexedBoundaryLocalizedRefinedPartition.representedStokes_of_zeroLocalStokesData
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.
  hasIntrinsicRoute_of_selectedSmoothRefinement
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.hasIntrinsicRoute_intrinsic
CoverIndexedZeroCompactRepresentedStokesIntrinsicInput.representedStokes
```

C.8 Axiom audit command

The theorem-level audit command is:

```
#print axioms
  Stokes.CoverIndexedZeroCompactRepresentedStokesFirstPrinciplesInput.
    representedStokes
```

The expected axioms are:

`propext, Classical.choice, Quot.sound.`

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